Étale Cohomology

Séminaire de Géométrie Algébrique du Bois-Marie SGA $4\frac{1}{2}$

Pierre Deligne

In collaboration with J.F. Boutot, A. Grothendieck, L. Illusie, and J.L. Verdier

Translated by Ajay Srinivasan

October 11, 2026

Translator’s Preface

...

Introduction

This volume aims to facilitate the use of ℓ-adic cohomology for the non-expert. I hope that it will often allow them to avoid resorting to the terse exposés in SGA 4 and SGA 5. It also contains some new results.

The first exposé, written by J.F. Boutot, surveys SGA 4. It provides the principal results - with a minimal generality, often insufficient for applications - and an idea of their proof. For complete results, or detailed proofs, SGA 4 remains indispensible.

The "Report on the trace formula" contains a complete proof of the trace formula for the Frobenius endomorphism. The proof is that given by Grothendieck in SGA 5, stripped of the unnecessary details. This Report should allow the user to forget SGA 5, which we may consider as a series of digressions, some very interesting. Its existence allows the publishing of SGA 5 as is soon. It is completed with the exposé "Applications of the trace formula to trigonometric sums" which explains how the trace formula allows for the study of trigonometric sums, and provides examples.

The target audience of the other expositions is more limited, and their writing style reflects this. The exposé "L-functions modulo ℓ^n and modulo p" is a modular generalization of the Report, based on the study in SGA 4 XVII 5.5 of symmetric powers. The exposé "The cohomology class associated to a cycle" defines this class in various contexts, and provides the compatibility between intersections and cup-products. In "Duality" are gathered some known results that lack a reference, and some compatibilities. The exposé "Theorems of finiteness in ℓ-adic cohomology" is new. It notably provides, in cohomology without support, theorems of finiteness analogous to those known in cohomology with compact support.

For more details on the exposés, I refer to their respective introductions.

I finally thank J.L. Verdier for allowing me to reproduce here his notes "Derived Categories (État 0)." I believe they remain very useful, and had become untraceable.

Bures-sur-Yvette, 20 September 1976
Pierre Deligne

Contents

1 Ariadne's thread for SGA 4, SGA $4\frac{1}{2}$, and SGA 5 **7**

2 Étale cohomology : the starting points **11**
Written by J.F. Boutot

3 Report on the trace formula **63**

4 L-functions modulo ℓ^n and modulo p **65**

5 The cohomology class associated to a cycle **67**
By A. Grothendieck, written by P. Deligne

6 Duality **69**

7 Applications of the trace formula to trigonometric sums **71**

8 Theorems of finiteness in ℓ-adic cohomology **73**
With an appendix by L. Illusie

9 Derived categories, état 0 **75**
By J.L. Verdier

10 Erratum for SGA 4 **77**

Chapter 1

Ariadne's thread for SGA 4, SGA $4\frac{1}{2}$, and SGA 5

The exposés I to VI of SGA 4 provide the general theory of Grothendieck topologies. Very detailed, they can be valuable when studying exotic topologies, like the one that gave birth to crystalline cohomology. For the étale topology, which is so close to classical intuition, such an imposing guardrail is not necessary : it suffices to be familiar with (for example) the book of Godement [God58], and to have a bit of faith. Other possible references: chapters I to III of Artin's notes [Art62], the Bourbaki exposé of Giraud [Gir64] or 1 in this volume. The exposés VII and VIII commence the study of étale topology. They are more detailed than chapter III of Artin and than 2 here.

The study of curves is the key to étale cohomology. It is commenced by Artin in exposé IX of SGA 4; the consequences, as for the cohomological dimension, are given in X. The essential points are repeated in 3 here. Exposé XI is not necessary for the following; it contains an elegant proof of the comparison theorem between étale cohomology and classical cohomology in the particular case of smooth complex algebraic varieties.

Exposés XII and XIII prove the fundamental base-change theorem for a proper morphism. As Artin noted in [Art69], his approximation theorem allows one to simplify the proof. This path is followed in 4 where multiple dévissages of the proof are only sketched, and where the applications of exposé XIV are only very briefly indicated. The theorem allows one to define "cohomology with compact support," and the "higher direct image with proper support" functors $R^i f_!$ (the properties of which are reduced to those of higher direct image functors for a proper morphism). This is the subject of the verbose exposé XVII; the essential point is said in 4, but the pieces of XVII can be of independent utility.

Exposés XV and XVI are centered on the fundamental theorem of local acyclicity for smooth morphisms. The influence of SGA 7 shows through in the less-detailed parallel exposition in 5, which also contains the proof of required results from SGA 2.

Exposé 6 is dedicated to Poincaré duality; it is clearer than SGA 4 XVIII but does not pretend to give a complete proof. For those who care about a purely algebraic proof, it is simplified by a reference to 3; it does not contain the proof of the formalism that the trace morphism fits into (XVIII §2).

There exists in étale cohomology a formalism of duality analogous to that of coherent cohomology. To establish this, Grothendieck used the resolution of singularities and the purity conjecture (for the statement, see 5 2.1.4), established in a relative framework in SGA 4 XVI, and - modulo the resolution - in equal characteristic in SGA 4 XIX. The key-points are established by another method, in 8, for schemes of finite type over a regular scheme of dimension 0 or 1. Various developments are given in SGA 5 I. In SGA 5 III, we show how this formalism implies the very general trace formula of Lefschetz-Verdier.

We see that in the original version of SGA 5, the Lefschetz-Verdier formula was only established conjecturally. Moreover, the local terms were not calculated there. For an application to L-functions, this seminar contains another proof, it is complete, in the particular case of the Frobenius morphism. This is the one in 2. Other references: for the statement and the schema of dévissages: the Bourbaki exposé of Grothendieck [Gro95]; for a brief description of the reduction (due to Grothendieck) of the crucial case to a case already treated by Weil, [Del76] §10; for an ℓ-adic treatment of this last case, 3.

In SGA 5, we will again find a detailed treatment of the passing from $\mathbb{Z}/\ell^n$-cohomology to $\mathbb{Z}_\ell$-cohomology (exposés V and VI, by J.P. Jouanolou), and of the formalism of the Frobenius morphism (exposé XV, by C. Houzel). The exposés X and XII, written by I. Bucur, give the calculation of the Euler-Poincaré characteristic of a sheaf on a curve, and that of the trace of the endomorphism of its cohomology defined by certain correspondances. In VII, Jouanolou gives the calculation of the cohomology of classical schemes, and applications. Otherwise, Serre's presentation in "Introduction à la théorie de Brauer" was recalled in his book [Ser78] (3rd part).

References

[Art62] M. Artin. *Grothendieck Topologies: Notes on a Seminar*. Harvard University, Department of Mathematics, 1962. URL: https://books.google.co.in/books?id=wERAAQAAIAAJ.

[Art69] M. Artin. "Algebraic approximation of structures over complete local rings". In: *Inst. Hautes Études Sci. Publ. Math.* 36 (1969), pp. 23–58. ISSN: 0073-8301,1618-1913. URL: http://www.numdam.org/item?id=PMIHES_1969__36__23_0.

[Del76] Pierre Deligne. "Les constantes locales de l'équation fonctionnelle de la fonction L d'Artin d'une représentation orthogonale". In: *Invent. Math.* 35 (1976), pp. 299–316. ISSN: 0020-9910,1432-1297. DOI: 10.1007/BF01390143. URL: https://doi.org/10.1007/BF01390143.

[Gir64] Jean Giraud. "Analysis situs". In: *Séminaire Bourbaki, 1962/63. Fasc. 3*. Secrétariat mathématique, Paris, 1964, No. 256, 11.

[God58] Roger Godement. *Topologie algébrique et théorie des faisceaux*. Vol. No. 13. Publications de l'Institut de Mathématiques de l'Université de Strasbourg [Publications of the Mathematical Institute of the University of Strasbourg]. Actualités Scientifiques et Industrielles, No. 1252. [Current Scientific and Industrial Topics]. Hermann, Paris, 1958, pp. viii+283.

[Gro95] Alexander Grothendieck. "Formule de Lefschetz et rationalité des fonctions L". In: *Séminaire Bourbaki, Vol. 9*. Soc. Math. France, Paris, 1995, Exp. No. 279, 41–55. ISBN: 2-85629-042-6.

[Ser78] Jean-Pierre Serre. *Représentations linéaires des groupes finis*. revised. Hermann, Paris, 1978, p. 182. ISBN: 2-7056-5630-8.

[SGA2] Alexander Grothendieck. *Cohomologie locale des faisceaux cohérents et théorèmes de Lefschetz locaux et globaux (SGA 2)*. Ed. by Yves Laszlo. Vol. 4. Documents Mathématiques (Paris) [Mathematical Documents (Paris)]. Séminaire de Géométrie Algébrique du Bois Marie, 1962., Augmenté d'un exposé de Michèle Raynaud. [With an exposé by Michèle Raynaud], Revised reprint of the 1968 French original. Société Mathématique de France, Paris, 2005, pp. x+208. ISBN: 2-85629-169-4.

[SGA4V1] *Théorie des topos et cohomologie étale des schémas. Tome 1: Théorie des topos*. Vol. Vol. 269. Lecture Notes in Mathematics. Séminaire de Géométrie Algébrique du Bois-Marie 1963–1964 (SGA 4), Dirigé par M. Artin, A. Grothendieck, et J. L. Verdier. Avec la collaboration de N. Bourbaki, P. Deligne et B. Saint-Donat. Springer-Verlag, Berlin-New York, 1972, pp. xix+525.

[SGA4V2] *Théorie des topos et cohomologie étale des schémas. Tome 2*. Vol. Vol. 270. Lecture Notes in Mathematics. Séminaire de Géométrie Algébrique du Bois-Marie 1963–1964 (SGA 4), Dirigé par M. Artin, A. Grothendieck et J. L. Verdier. Avec la collaboration de N. Bourbaki, P. Deligne et B. Saint-Donat. Springer-Verlag, Berlin-New York, 1972, pp. iv+418.

[SGA4V3] *Théorie des topos et cohomologie étale des schémas. Tome 3*. Vol. Vol. 305. Lecture Notes in Mathematics. Séminaire de Géométrie Algébrique du Bois-Marie 1963–1964 (SGA 4), Dirigé par M. Artin, A. Grothendieck et J. L. Verdier. Avec la collaboration de P. Deligne et B. Saint-Donat. Springer-Verlag, Berlin-New York, 1973, pp. vi+640.

[SGA5] *Cohomologie l-adique et fonctions* L. Vol. Vol. 589. Lecture Notes in Mathematics. Séminaire de Géometrie Algébrique du Bois-Marie 1965–1966 (SGA 5), Edité par Luc Illusie. Springer-Verlag, Berlin-New York, 1977, pp. xii+484. ISBN: 3-540-08248-4.

Chapter 2

Étale cohomology : the starting points

by P. Deligne, written by J.F. Boutot

This work is based on 6 lectures given by P. Deligne at Arcata in August 1974 (AMS Summer School on algebraic geometry), under the title: "Inputs of etale cohomology." A 7th-presentation became the "report on the trace formula," in this volume. The goal of these talks was to give proofs of fundamental theorems in étale cohomology, freed from the gangue of nonsense that surrounds them in SGA 4. We have not sought to state the theorems under their most general form, nor followed the dévissages, sometimes clever, that their proof required. We have on the contrary placed the emphasis on the "irreducible" cases, which, with all dévissages done, remained to be treated.

We hope that this text, which does not claim any originality, aids the reader to consult with profit the 3 volumes of SGA 4.

Convention 0.0.1. We only consider quasi-compact (= union of finite affine open subschemes) and quasi-separated (= such that the intersection of any two open affine subschemes is quasi-compact) schemes, and we simply call them schemes.

Contents

1 Grothendieck topologies . . . 13
1.1 Sieves . . . 13
1.2 Sheaves . . . 13
1.3 Bundles . . . 14
1.4 Faithfully flat descent . . . 15
1.5 A particular case : Hilbert's theorem 90 . . . 18
1.6 Grothendieck topologies . . . 19
2 Étale topology . . . 22
2.1 Étale topology . . . 22
2.2 Examples of sheaves . . . 23
2.3 Fibers, direct images . . . 25
2.4 Galois cohomology . . . 27
3 Cohomology of curves . . . 29
3.1 Brauer group . . . 29
3.2 Tsen's theorem . . . 31
3.3 Cohomology of smooth curves . . . 32
3.4 Dévissages . . . 35
4 Base-change theorem for a proper morphism . . . 37
4.1 Introduction . . . 37
4.2 Proof for $q = 0$ or 1 and $\mathcal{F} = \mathbb{Z}/n$ . . . 37
4.3 Constructible sheaves . . . 39
4.4 End of the proof . . . 41
4.5 Cohomology with proper support . . . 42
4.6 Applications . . . 44
5 Local acyclicity of smooth morphisms . . . 45
5.1 Locally acyclic morphisms . . . 46
5.2 Local acyclicity of a smooth morphism . . . 49
5.3 Applications . . . 51
6 Poincaré Duality . . . 54
6.1 Introduction . . . 54
6.2 The case of curves . . . 55
6.3 The general case . . . 57
6.4 Variants and Applications . . . 58
References . . . 60

1 Grothendieck topologies

At their origin, Grothendieck topologies appeared as underlying his theory of descent (cf. [SGA1, pp. VI, VIII]); the usage of corresponding cohomology theories is later. The same approach is taken here in formalizing the classical notions of localization, of the local property of gluing, we bring out the general concept of the Grothendieck topology; to justify its introduction into algebraic geometry, we show a theorem of faithfully flat descent, a generalization of the classical Hilbert's theorem 90.

1.1 Sieves

Let X be a topological space, and $f : X \to \mathbb{R}$ a real-valued function on X. The continuity of f is a property of local nature; in other words, if f is continuous on every sufficiently small open subset of X, f is continuous on all of X. To formalize the notion of a "property of local nature," we introduce some definitions.

We say that a set $\mathcal{U}$ of open subsets of X is a *sieve* if for every $U \in \mathcal{U}$ and $V \subset U$, we have $V \in \mathcal{U}$. We say that a sieve is *covering* if the union of all of the open subsets belonging to this sieve is equal to X.

Given a family $\{U_i\}_{i\in J}$ of open subsets of X, the sieve generated by $\{U_i\}_{i\in J}$ is by definition the set of open subsets U of X such that U is contained in some U_i.

We say that a property $P(U)$, defined for every open subset U of X, is *local* if, for every covering sieve $\mathcal{U}$ of every open subset U of X, $P(U)$ is true if and only if $P(V)$ is true for every $V \in \mathcal{U}$. For example, given $f : X \to \mathbb{R}$, the property "f is continuous on U" is local.

1.2 Sheaves

We make precise the notion of locally given functions on X.

1.2.1. **Point of view of sieves**: Let $\mathcal{U}$ be a sieve of open subsets of X. We mean by a $\mathcal{U}$-locally given function on X the data for every $U \in \mathcal{U}$ of a function f_U on U such that, if $V \subset U$, we have $f_V = f_U|_V$.

1.2.2. **Point of view of Čech**: If the sieve $\mathcal{U}$ is generated by a family of open subsets U_i of X, to provide a function $\mathcal{U}$-locally is to provide a function f_i on each U_i such that $f_i|_{U_i\cap U_j} = f_j|_{U_i\cap U_j}$.

Said differently, if $Z = \coprod U_i$, giving a function $\mathcal{U}$-locally is to provide a function on Z which is constant on the fibers of the natural projec-

tion $Z \to X$.

1.2.3. Continuous functions form a sheaf; this means that for every covering sieve $\mathcal{U}$ of an open subset V of X and every $\mathcal{U}$-locally given function $\{f_U\}$ such that every f_U is continuous on U, there exists a unique continuous function f on V such that $f|_U = f_U$ for every $U \in \mathcal{U}$.

1.3 Bundles

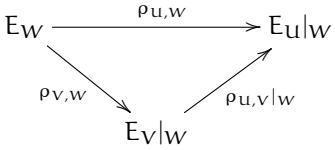
We will make precise now the notion of a locally given vector bundle on X.

1.3.1. **Point of view of sieves:** Let $\mathcal{U}$ be a sieve of open subsets of X. We mean by a $\mathcal{U}$-locally given vector bundle on X the data of

a) a vector bundle E_U over each $U \in \mathcal{U}$.

b) if $V \subset U$, an isomorphism $\rho_{U,V} : E_V \xrightarrow{\sim} E_U|_V$, such that the following is true.

c) if $W \subset V \subset U$, the diagram



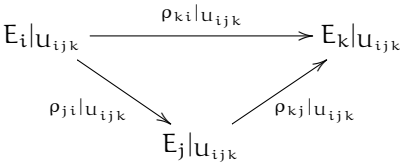
commutes, i.e. $\rho_{U,W} = (\rho_{U,V}|_W) \circ \rho_{V,W}$.

1.3.2. **Point of view of Čech:** If the sieve $\mathcal{U}$ is generated by a family of open subsets U_i of X, providing a vector bundle $\mathcal{U}$-locally is to provide:

a) a vector bundle E_i on each U_i,

b) if $U_{ij} = U_i \cap U_j = U_i \times_X U_j$, an isomorphism $\rho_{ji} : E_i|_{U_{ij}} \xrightarrow{\sim} E_j|_{U_{ij}}$, so that

c) if $U_{ijk} = U_i \times_X U_j \times_X U_k$, the diagram



commutes, i.e. $\rho_{ki} = \rho_{kj} \circ \rho_{ji}$ on U_{ijk}.

Said differently, if $Z = \coprod U_i$ and if $\pi : Z \to X$ is the natural projection, to provide a vector bundle $\mathcal{U}$-locally is to provide:

a) a vector bundle E over Z,

b) if x and y are two points in Z such that $\pi(x) = \pi(y)$, an isomorphism $\rho_{yx} : E_x \xrightarrow{\sim} E_y$ between the fibers of E at x and at y, depending continuously on (x, y) and such that,

c) if x, y and z are three points of Z such that $\pi(x) = \pi(y) = \pi(z)$, we have $\rho_{zx} = \rho_{zy} \circ \rho_{yx}$.

1.3.3. A vector bundle E over X defines a vector bundle given $\mathcal{U}$-locally $E_{\mathcal{U}}$: the system of restrictions E_U of E to objects of $\mathcal{U}$. The fact that the notion of a vector bundle is of local nature may be expressed so: for every covering sieve $\mathcal{U}$ of X, the functor $E \mapsto E_{\mathcal{U}}$, is an equivalence of categories.

1.3.4. If in 1.1, we replace "open subset of X" by "subspace of X," we obtain the notion of a *sieve of subspaces of* X. In this framework too we have gluing theorems. For example: let X be a normal space and $\mathcal{C}$ a sieve of subspaces of X generated by a locally finite closed cover of X, then the functor $E \mapsto E_{\mathcal{C}}$, from vector bundles over X to $\mathcal{C}$-locally given vector bundles is an equivalence of categories.

In algebraic geometry, it is useful to also consider "sieves of spaces over X." This is what we will see in the following paragraph.

1.4 Faithfully flat descent

1.4.1. In the framework of schemes, the Zariski topology is not fine enough for the study of non-linear problems and we are brought to replace in the preceding definitions, open immersions by more general morphisms. From this point of view, the techniques of descent appear as localization techniques. So the following statement of descent can be expressed by saying that the considered properties are of local nature for the faithfully flat topology (we say that a morphism of schemes is faithfully flat if it is flat and surjective).

Proposition 1.4.2. *Let A be a ring and B a faithfully flat A-algebra. Then:*

(i) *A sequence $\Sigma = (M' \longrightarrow M \longrightarrow M'')$ of A-modules is exact when the sequence $\Sigma_{(B)}$ deduced by extension of scalars to B is exact.*

(ii) *An A-module M is of finite type (resp. of finite presentation, flat, locally free of finite rank, invertible (i.e. locally free of rank 1)) when the B-module $M_{(B)}$ is.*

Proof. With the functor $M \mapsto M_{(B)}$ being exact (flatness of B), it suffices to show that, if an A-module N is nonzero, $N_{(B)}$ is nonzero. If N is nonzero,

N contains a nonzero cyclic submodule A/a; so $N_{(B)}$ contains a cyclic submodule $(A/a)_{(B)} = B/aB$, nonzero by surjectivity of the structural morphism $\varphi : \mathrm{Spec}(B) \longrightarrow \mathrm{Spec}(A)$ (if $V(a)$ is nonempty, $\varphi^{-1}(V(a)) = V(aB)$ is nonempty).

For every family (x_i) of elements of $M_{(B)}$, there exists a submodule of finite type M' of M such that $M'_{(B)}$ contains the x_i's. If $M_{(B)}$ is of finite type and if the x_i's generate $M_{(B)}$, we have $M'_{(B)} = M_{(B)}$, so $M' = M$ and M is of finite type.

If $M_{(B)}$ is of finite presentation, we can, from above, find a surjection $A^n \longrightarrow M$. If N is the kernel of this surjection, the B-module $N_{(B)}$ is of finite type, so N is, and M is of finite presentation. The assertion for "flat" results also from (i); "locally free of finite rank" means "flat and of finite presentation" and the rank is tested by extension of scalars to fields.

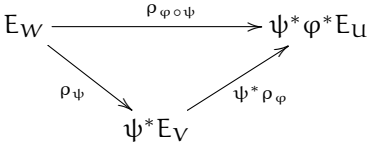
1.4.3. Let X be a scheme and $\mathscr{S}$ a class of X-schemes stable under fiber product over X. A class $\mathcal{U} \subset \mathscr{S}$ is a *sieve* on X (relative to $\mathscr{S}$) if, for every morphism $\varphi\colon V \to U$ of X-schemes, with $U, V \in \mathscr{S}$ and $U \in \mathcal{U}$, we have $V \in \mathcal{U}$. The sieve *generated* by a famillly $\{U_i\}_{i\in I}$ of X-schemes in $\mathscr{S}$ is the class of $V \in \mathscr{S}$ such that there exists a morphism of X-schemes from V into some U_i.

1.4.4. Let $\mathcal{U}$ be a sieve on X. We mean by a $\mathcal{U}$-locally given quasi-coherent module on X, the data of:

(a) a quasi-coherent module E_U on each $U \in \mathcal{U}$,

(b) for every $U \in \mathcal{U}$ and for every morphism $\varphi\colon V \to U$ of X-schemes in $\mathscr{S}$, an isomorphism $\rho_\varphi : E_V \xrightarrow{\sim} \varphi^* E_U$, such that

(c) if $\psi\colon W \longrightarrow V$ is a morphism of X-schemes in $\mathscr{S}$, the diagram



commutes, i.e. $\rho_{\varphi\circ\psi} = (\psi^*\rho_\varphi) \circ \rho_\psi$.

If E is a quasi-coherent module on X, we denote by $E_{\mathcal{U}}$ the $\mathcal{U}$-locally given module having value $\varphi_U^* E$ on $\varphi_U : U \longrightarrow X$ and such that, for every morphism $\psi\colon V \longrightarrow U$ is the restriction isomorphism ρ_ψ is the canonical isomorphism $E_V = (\varphi_U \circ \psi)^* E \xrightarrow{\sim} \psi^* \varphi_U^* E = \psi^* E_U$.

Theorem 1.4.5. *Let $\{U_i\}_{i\in J} \in \mathscr{S}$ a finite family of flat X-schemes on X such that X is the union of the images of U_i, and let $\mathcal{U}$ be the sieve generated by $\{U_i\}_{i\in J}$. Then the functor $E \mapsto E_{\mathcal{U}}$ is an equivalence from the category of quasi-coherent modules on X to the category of $\mathcal{U}$-locally given quasi-coherent modules*

Proof. We only treat the case where X is affine and where $\mathcal{U}$ is generated by an affine X-scheme U, faithfully flat on X. The reduction to this case is formal. We let $X = \mathrm{Spec}(A)$ and $U = \mathrm{Spec}(B)$.

If the morphism $U \longrightarrow X$ admits a section, X belongs to the sieve $\mathcal{U}$ and the assertion is evident. We reduce to this case.

A $\mathcal{U}$-locally given quasi-coherent module defines modules M', M'', M''' on U, $U \times_X U$ and $U \times_X U \times_X U$, and isomorphisms $\rho\colon p^*M^\bullet \simeq M^\bullet$ for every projection morphism p between these spaces; this is a cartesian diagram

$$M^*\colon M' \rightrightarrows M'' \;\substack{\rightarrow\\ \rightarrow\\ \rightarrow}\; M'''$$

over

$$U_*\colon U \;\substack{\leftarrow\\ \leftarrow}\; U \times_X U \;\substack{\leftarrow\\ \leftarrow\\ \leftarrow}\; U \times_X U \times_X U$$

Conversely, M^* determines the $\mathcal{U}$-locally given module: for $V \in \mathcal{U}$, there exists $\varphi : V \longrightarrow U$ and we write $M_U = \varphi^*M'$; for $\varphi_1, \varphi_2\colon V \longrightarrow U$, we have a natural identification $\varphi_1^*M' \simeq (\varphi_1 \times \varphi_2)^*M'' \simeq \varphi_2^*M'$, and we see using M''' that these identifications are compatible, so that the definition is legitimate. In short, it comes down to the same thing, to $\mathcal{U}$-locally give a module or a cartesian diagram M^* on U_*.

Translating to algebraic terms: to give M^* is to give a cartesian diagram of modules

$$M' \;\substack{\xrightarrow{\partial_0}\\ \xrightarrow{\partial_1}}\; M'' \;\substack{\xrightarrow{\partial_0}\\ \xrightarrow{\partial_1}\\ \xrightarrow{\partial_2}}\; M'''$$

over the diagram of rings

$$B \;\substack{\xrightarrow{\partial_0}\\ \xrightarrow{\partial_1}}\; B \otimes_A B \;\substack{\xrightarrow{\partial_0}\\ \xrightarrow{\partial_1}\\ \xrightarrow{\partial_2}}\; B \otimes_A B \otimes_A B$$

[to make that precise: we have $\partial_i(bm) = \partial_i(b) \cdot \partial_i(m)$, the usual identities such that $\partial_0\partial_1 = \partial_0\partial_0$ are true, and "cartesian" means that the morphisms $\partial_i\colon M' \otimes_{B,\partial_1} (B \otimes_A B) \longrightarrow M''$ and $M'' \otimes_{B\otimes_A B,\partial_i} (B \otimes_A B \otimes_A B) \longrightarrow M'''$ are isomorphisms].

The functor $E \mapsto E_{\mathcal{U}}$ becomes the functor which, to an A-module M, associates

$$M^* = (M \otimes_A B \rightrightarrows M \otimes_A B \otimes_A B \;\substack{\rightarrow\\ \rightarrow\\ \rightarrow}\; M \otimes_A B \otimes_A B \otimes_A B).$$

It admits for a right adjoint, the functor

$$(M' \rightrightarrows M'' \mathrel{\substack{\rightarrow\\[-1ex]\rightarrow\\[-1ex]\rightarrow}} M''') \mapsto \ker(M' \rightrightarrows M'').$$

We have to prove that the adjunction arrows:

$$M \longrightarrow \ker(M \otimes_A B \rightrightarrows M \otimes_A B \otimes_A B)$$

and

$$\ker(M' \rightrightarrows M'') \otimes_A B \longrightarrow M'$$

are isomorphisms. According to Prop. 1.4.2(i), it suffices to just prove it after a faithfully flat base change $A \longrightarrow A'$ (B becomes $B' = B \otimes_A A'$). Taking $A' = B$, this brings us back to the case where $U \longrightarrow X$ admits a section.

1.5 A particular case : Hilbert's theorem 90

1.5.1. Let k be a field, k' a Galois extension of k and $G = \mathrm{Gal}(k'/k)$. Then the homomorphism

$$k' \otimes_k k' \longrightarrow \bigoplus_{\sigma \in G} k'$$
$$x \otimes y \mapsto \{x \cdot \sigma(y)\}_{\sigma \in G}$$

is bijective.

We deduce from this that to locally provide a module for a sieve generated by $\mathrm{Spec}(k')$ over $\mathrm{Spec}(k)$ is to provide a vector k'-space equipped with a semi-linear G-action, i.e.:

a) a vector k'-space V',

b) for every $\sigma \in G$, an endomorphism φ_σ of the group structure of V' such that $\varphi_\sigma(\lambda v) = \sigma(\lambda)\varphi_\sigma(v)$, for every $\lambda \in k'$ and $v \in V'$, satisfying the condition

c) for every $\sigma, \tau \in G$, we have $\varphi_{\tau\sigma} = \varphi_\tau \circ \varphi_\sigma$.

Let $V = V'^G$ be the group of invariants by this G-action; this is a vector K-space, according to Thm. 1.4.5, we have:

Proposition 1.5.2. *The inclusion of V in V' defines an isomorphism $V \otimes_K K' \xrightarrow{\sim} V'$.*

In particular, if V' is of dimension l and if $v' \in V$ is nonzero, φ_σ and determined by the constant $c(\sigma) \in K'^*$ such that $\varphi_\sigma(v') = c(\sigma)v'$ and the condition c) is written as

$$c(\tau\sigma) = c(\tau) \cdot \tau(c(\sigma)).$$

according to the proposition there exists a nonzero invariant vector $v = \mu v'$, $\mu \in k'^*$. We therefore have for every $\sigma \in G$,

$$c(\sigma) = \mu \cdot \sigma(\mu^{-1}).$$

Said otherwise, every 1-cocycle of G with values in k'^* is a cobordism.

Corollary 1.5.3. *We have that* $H^1(G, k'^*) \cong 0$.

1.6 Grothendieck topologies

We transcribe now the definitions of the previous paragraphs in a global abstract framework for both the case of topological spaces and that of schemes.

1.6.1. Let $\mathscr{S}$ be a category and U an object of $\mathscr{S}$. We mean by a *sieve* on U a subset $\mathcal{U}$ of $\mathrm{Ob}(\mathscr{S}/U)$ such that if $\varphi\colon V \longrightarrow U$ belongs to $\mathscr{U}$ and if $\psi\colon W \longrightarrow V$ is a morphism in $\mathscr{S}$, so $\varphi \circ \psi\colon W \longrightarrow U$ belongs to $\mathcal{U}$.

If $\{\varphi_i\colon U_i \longrightarrow U\}$ is a family of morphisms, the sieve generated by the U_i is by definition the set of morphisms $\varphi\colon V \longrightarrow U$ that factorize through one of the φ_i.

If $\mathcal{U}$ is a sieve on U and if $\varphi\colon V \longrightarrow U$ is a morphism, the restriction $\mathcal{U}_V$ of $\mathcal{U}$ to V is by definition the sieve on V made up of the morphisms $\psi\colon W \longrightarrow V$ such that $\varphi \circ \psi\colon W \longrightarrow U$ belongs to $\mathcal{U}$.

1.6.2. The data of a *Grothendieck topology* on $\mathscr{S}$ consists of the data for every object U of $\mathscr{S}$, a set $C(U)$ of sieves on U, called covering sieves, such that the following axioms are satisfied:

a) The sieve generated by the identity of U is covering.

b) If $\mathcal{U}$ is a covering sieve on U and if $V \longrightarrow U$ is a morphism, the sieve $\mathcal{U}_V$ is covering.

c) A locally covering sieve is covering. Said otherwise, if $\mathcal{U}$ is a covering sieve on U and if $\mathcal{U}'$ is a sieve on U such that, for every $V \longrightarrow U$ belongs to $\mathcal{U}$, the sieve $\mathcal{U}'_V$ is covering, then $\mathcal{U}'$ is covering.

We mean by a *site* the data of a category equipped with a Grothendieck topology.

1.6.3. Given a site $\mathscr{S}$, we call a contravariant functor $\mathcal{F}$ from $\mathscr{S}$ to the category of sets a *presheaf* on $\mathscr{S}$. For every object U of $\mathscr{S}$, we call the elements of $\mathcal{F}(U)$ the sections of $\mathcal{F}$ over U. For every morphism $V \longrightarrow U$

and for every $s \in \mathcal{F}(U)$, we denote by $s|_V$ (s restricted to V) the image of s in $\mathcal{F}(V)$.

If $\mathcal{U}$ is a sieve on U, we mean by a $\mathcal{U}$-locally given section the data, for every $V \longrightarrow U$ in $\mathcal{U}$, of a section $s_V \in \mathcal{F}(V)$ such that, for every morphism $W \longrightarrow V$, we have $s_V|_W = s_W$. We say that $\mathcal{F}$ is a *sheaf* if, for every object U of $\mathscr{S}$, for every covering sieve $\mathcal{U}$ on U and for every $\mathcal{U}$-locally given section $\{s_V\}$, there exists a unique section $s \in \mathcal{F}(U)$ such that $s|_V = s_V$, for every $V \longrightarrow U$ in $\mathcal{U}$.

We define abelian sheaves in an analogous manner by replacing the category of sets by that of abelian groups. We can show that the category of abelian sheaves on $\mathscr{S}$ is an abelian category with sufficiently many injectives. A sequence $\mathcal{F} \xrightarrow{f} \mathcal{G} \xrightarrow{g} \mathcal{H}$ of sheaves is exact if, for every object U of $\mathscr{S}$ and for every $s \in \mathcal{G}(U)$ such that $g(s) = 0$, there locally exists t such that $f(t) = s$; i.e. if there exists a covering sieve $\mathcal{U}$ on U and for every $V \in \mathcal{U}$, a section t_V of $\mathcal{F}$ over V such that $f(t_V) = s|_V$.

1.6.4. **Examples**: We saw two above.

a) Let X be a topological space and $\mathscr{S}$ the category whose objects are open subsets of X and whose morphisms are natural inclusions. The Grothendieck topology on $\mathscr{S}$ corresponding to the usual topology of X is that for which a sieve $\mathcal{U}$ on an open subset U of X is covering if the union of the open subsets in this is sieve is equal to U. It is clear that the category of sheaves on $\mathscr{S}$ is equivalent to the category of sheaves on X in the usual sense.

b) Let X be a scheme and $\mathscr{S}$ the category of schemes over X. We mean by the qcff (quasi-compact faithfully flat) topology on $\mathscr{S}$ the Grothendieck topology for which a sieve on an X-scheme U is covering if it is generated by a finite family of flat morphisms whose images cover U.

1.6.5. **Cohomology:** We always suppose that the category $\mathscr{S}$ has a final object X. So we mean by the global sections of an abelian sheaf $\mathcal{F}$, and we denote by $\Gamma\mathcal{F}$ or $H^0(X, \mathcal{F})$, the group $\mathcal{F}(X)$. The functor $\mathcal{F} \mapsto \Gamma\mathcal{F}$ is a left exact functor from the category of abelian sheaves on $\mathscr{S}$ to the category of abelian groups, and we denote by $H^i(X, \bullet)$ its derived functors (or satellites). These cohomology groups represent the obstructions to passing from the local to the global. By definition, if $0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{G} \longrightarrow \mathcal{H} \longrightarrow 0$ is an exact sequence of abelian sheaves, we have a long exact sequence in cohomology:

$$0 \longrightarrow H^0(X,\mathcal{F}) \longrightarrow H^0(X,\mathcal{G}) \longrightarrow H^0(X,\mathcal{H}) \longrightarrow H^1(X,\mathcal{F}) \longrightarrow \cdots$$
$$\cdots \longrightarrow H^n(X,\mathcal{F}) \longrightarrow H^n(X,\mathcal{G}) \longrightarrow H^n(X,\mathcal{H}) \longrightarrow H^{n+1}(X,\mathcal{F}) \longrightarrow \cdots$$

1.6.6. Given an abelian sheaf $\mathcal{F}$ on $\mathscr{S}$, we mean by an $\mathcal{F}$-torsor a sheaf $\mathcal{G}$ equipped with an action $\mathcal{F} \times \mathcal{G} \longrightarrow \mathcal{G}$ of $\mathcal{F}$ such that locally (after restriction to all objects of a sieve covering the final object X) $\mathcal{G}$ equipped with the action of $\mathcal{F}$ is isomorphic to $\mathcal{F}$ equipped with the canonical action $\mathcal{F} \times \mathcal{F} \longrightarrow \mathcal{F}$ by translations.

We can show that $H^i(X,\mathcal{F})$ is interpreted as the set of isomorphism classes of $\mathcal{F}$-torsors.

2 Étale topology

We specialize the definitions of the preceding chapter to the case of the étale topology of a scheme X (2.1, 2.2, 2.3). The corresponding cohomology coincides in the case where X is the spectrum of a field K with the Galois cohomology of K (2.4).

2.1 Étale topology

We start with some reminders on the notion of étale morphisms.

Definition 2.1.1. Let A be a (commutative) ring. We say that an A-algebra B is *étale* if B is an A-algebra of finite presentation and if the following equivalent conditions are satisfied:

a) For every A-algebra C and for every ideal of square zero J of C, the canonical map

$$\mathrm{Hom}_{A-\mathrm{alg}}(B, C) \longrightarrow \mathrm{Hom}_{A-\mathrm{alg}}(B, C/J)$$

is a bijection.

b) B is a flat A-module and $\Omega_{B/A} = 0$ (we denote by $\Omega_{B/A}$ the module of relative differentials).

c) Let $B = A[x_1, \cdots, x_n]/I$ be a presentation of B. Then for every prime ideal r of $A[x_1, \cdots, x_n]$ containing I, there exists polynomials $p_1, \cdots, p_n \in I$ such that I_r is generated by the images of $p_1, \cdots, p_n$ and $\det(\partial p_i/\partial x_i) \notin r$.

[cf. [SGA1, Exposé I] or [Ray70, Chapter V]].

We say that a morphism of schemes $f\colon X \longrightarrow S$ is étale if for every $x \in X$ there exists an affine open neighborhood $U = \mathrm{Spec}(A)$ of $f(x)$ and an affine open neighborhood $V = \mathrm{Spec}(B)$ of x in $X \times_S U$ such that B is an étale A-algebra.

2.1.2. **Examples:** a) If A is a field, an A-algebra B is étale if and only if it is a finite product of separable extensions of A.
b) If X and S are schemes of finite type over $\mathbb{C}$, a morphism $f\colon X \longrightarrow S$ is étale if and only if its analytification $f^{an}\colon X^{an} \longrightarrow S^{an}$ is a local isomorphism.

Sorite 2.1.3. a) (base-change) If $f\colon X \longrightarrow S$ is an étale morphism, the same is true of $f_{S'}\colon X \times_S S' \longrightarrow S'$ for every morphism $S' \longrightarrow S$.
b) (composition) The composite of two étale morphisms is an étale morphism.

c) If $f\colon X \longrightarrow S$ and $g\colon Y \longrightarrow S$ are two étale morphisms, every S-morphism from X to Y is étale.

d) (descent) Let $f\colon X \longrightarrow S$ be a morphism. If there exists a faithfully flat morphism $S' \longrightarrow S$, such that $f_{S'}\colon X \times_S S' \longrightarrow S'$ is étale, then f is étale.

2.1.1. Let X be a scheme. Let $\mathscr{S}$ the category of étale X-schemes; by 2.1.3 c) every morphism of $\mathscr{S}$ is an étale morphism. We mean by the *étale topology* on $\mathscr{S}$ the topology for which a sieve on U is covering if it is generated by a finite family of morphisms $\varphi_i\colon U_i \longrightarrow U$ such that the union of the images of φ_i covers U. We mean by the *étale site* of X, and we denote by $X_{\text{ét}}$, the site defined by $\mathscr{S}$ equipped with the étale topology.

2.2 Examples of sheaves

2.2.1. **Constant sheaf:** Let C be an abelian group and suppose for simplicity that X is Noetherian. We will note by C_X (or even C if there is no ambiguity) the sheaf defined by $U \mapsto C^{\pi_0(U)}$, where $\pi_0(U)$ is the (finite) set of connected components of U. The most important case will be $C = \mathbb{Z}/n$. We thus have by definition

$$H^0(X, \mathbb{Z}/n) = \mathbb{Z}/n^{\pi_0(X)}.$$

Moreover $H^1(X, \mathbb{Z}/n)$ is the set of isomorphism classes of $\mathbb{Z}/n$-torsors (1.6.6.), in other words the Galois étale coverings of X with group $\mathbb{Z}/n$. In particular, if X is connected and if $\pi_1(X)$ is its fundamental group for a chosen basepoint, we have

$$H^1(X, \mathbb{Z}/n) = \mathrm{Hom}(\pi_1(X), \mathbb{Z}/n).$$

2.2.2. **Multiplicative group:** We will note by $\mathbb{G}_{m,X}$ (or $\mathbb{G}_m$ if there is no ambiguity) the sheaf defined by $U \mapsto \Gamma(U, \mathcal{O}_U^*)$; it is indeed a sheaf thanks to the theorem of faithfully flat descent (1.4.5). We have by definition

$$H^0(X, \mathbb{G}_m) = H^0(X, \mathcal{O}_X)^*;$$

particularly if X is reduced, connected and proper over an algebraically closed K, we have:

$$H^0(X, \mathbb{G}_m) = K^*$$

Proposition 2.2.3. *We have an isomorphism:*

$$H^1(X, \mathbb{G}_m) = \mathrm{Pic}(X),$$

where $\mathrm{Pic}(X)$ *is the group of classes of invertible sheaves on* X.

Proof. Let $*$ be the functor that, to an invertible sheaf $\mathcal{L}$ on X, associates the following presheaf $\mathcal{L}^*$ on $X_{\text{ét}}$: for $\varphi\colon U \longrightarrow X$ étale,

$$\mathcal{L}^*(U) = \mathrm{Isom}_U(\mathcal{O}_U, \varphi^*\mathcal{L}).$$

By Prop. 1.4.2 (i) and Thm. 1.4.5 (fully faithful), this presheaf is a sheaf; it's also a $\mathbb{G}_m$-torsor. We also verify that

a) the functor $*$ is compatible with the localization (étale);

b) it induces an equivalence from the category of trivial invertible sheaves (i.e. isomorphic to $\mathcal{O}_X$) with the category of trivial $\mathbb{G}_m$-torsors: $\mathcal{L}$ is trivial iff $\mathcal{L}^*$ is.

Moreover, by Prop. 1.4.2 (ii) and Thm. 1.4.5,

c) the notion of an invertible sheaf is local for the étale topology.

It formally follows from a), b), c) that $*$ is an equivalence between the category of invertible sheaves on X and that of $\mathbb{G}_m$-torsors on $X_{\text{ét}}$; it induces the needed isomorphism. We construct the inverse equivalence as follows: if T is a $\mathbb{G}_m$-torsor, there exists a finite étale cover $\{U_i\}$ of X such that the torsors T/U_i are trivial; so T is trivial over each étale V on X in the sieve $\mathcal{U} \subset X_{\text{ét}}$ generated by $\{U_i\}$. On each $V \in \mathcal{U}$, $T|_V$ corresponds to an invertible sheaf $\mathcal{L}_V$ (by b)) and the $\mathcal{L}_V$ constitute a $\mathcal{L}_U$ (by a)). By c), the latter provides an invertible sheaf $\mathcal{L}(T)$ on X, and $T \mapsto \mathcal{L}(T)$ is the required inverse of $*$.

2.2.4. **Roots of unity:** For every integer $n > 0$, we mean by the sheaf of nth roots of unity, and we denote by μ_n, the kernel of raising to the nth power in $\mathbb{G}_m$. If X is a scheme on a separable algebraically closed k and if n is invertible in k, the choice of a primitive nth root of unity $\xi \in k$ defines an isomorphism $i \mapsto \xi^i$ of $\mathbb{Z}/n$ with μ_n.

The relation between cohomology with coefficients in μ_n and cohomology with coefficients in $\mathbb{G}_m$ is given by the exact sequence in cohomology deduced from the

Theorem 2.2.5 (Kummer Theory)**.** *If n is invertible on X, raising to the nth power in $\mathbb{G}_m$ is an epimorphism of sheaves. We then have an exact sequence*

$$0 \longrightarrow \mu_n \longrightarrow \mathbb{G}_m \longrightarrow \mathbb{G}_m \longrightarrow 0.$$

Proof. Let $U \longrightarrow X$ be an étale morphism and $a \in \mathbb{G}_m(U) = \Gamma(U, \mathcal{O}_U^*)$. Since n is invertible on U, the equation $T^n - a = 0$ is separable; said otherwise $U' = \mathrm{Spec}\, \mathcal{O}_U[T]/(T^n - a)$ is étale over U. Moreover $U' \longrightarrow U$ is surjective and admits an nth root on U', and the result follows.

2.3 Fibers, direct images

2.3.1. We call a *geometric point* of X a morphism $\overline{x} \longrightarrow X$, where $\overline{x}$ is the spectrum of a separably closed field $k(\overline{x})$. We will denote by $\overline{x}$ (by abuse of notation), underlying the morphism $\overline{x} \longrightarrow X$. If x is the image of $\overline{x}$ in X, we say that $\overline{x}$ is centered at x. If the field $k(\overline{x})$ is an algebraic extension of the residual field $k(x)$, we say that $\overline{x}$ is an *algebraic* geometric point of X.

We mean by an *étale neighborhood* of $\overline{x}$ a commutative diagram

$$\begin{array}{ccc} & & U \\ & \nearrow & \downarrow \\ \overline{x} & \longrightarrow & X \end{array}$$

where $U \longrightarrow X$ is an étale morphism.

The *strict localization* of X at $\overline{x}$ is the ring $\mathcal{O}_{X,\overline{x}} = \varinjlim \Gamma(U, \mathcal{O}_U)$, the inductive limit being taken over the étale neighborhoods of $\overline{x}$. This is a strictly henselian local ring whose residual field is the separable closure of the residual field $k(x)$ of X in x in $k(\overline{x})$. It plays the role of a local ring for the étale topology.

2.3.2. Given a sheaf $\mathcal{F}$ on $X_{\text{ét}}$, we mean by the *fiber* of $\mathcal{F}$ at $\overline{x}$ the set (resp. the group, $\cdots$) $\mathcal{F}_{\overline{x}} = \varinjlim \mathcal{F}(U)$, the inductive limit is always taken over the étale neighborhoods of $\overline{x}$.

For a homomorphism of sheaves $\mathcal{F} \longrightarrow \mathcal{G}$ to be a mono/epi/isomorphism it is neccessary and sufficient that this be the case for the homomorphisms $\mathcal{F}_{\overline{x}} \longrightarrow \mathcal{G}_{\overline{x}}$ induces on the fibers at every geometric point of X. If X is of finite type on an algebraically closed field, it suffices that it is so on the rational points of X.

2.3.3. If $f\colon X \longrightarrow Y$ is a morphism of schemes and F a sheaf on $X_{\text{ét}}$, the *direct image* $f_*\mathcal{F}$ of $\mathcal{F}$ by f is the sheaf on $Y_{\text{ét}}$ defined by $f_*\mathcal{F}(V) = F(X \times_Y V)$ for every V étale on Y.

The functor $f_*\colon \mathbf{Ab}(\mathbf{Sh}(X_{\text{ét}})) \longrightarrow \mathbf{Ab}(\mathbf{Sh}(Y_{\text{ét}}))$ is left exact. Its right derived functors $R^q f_*$ are called the higher direct images. If $\overline{y}$ is a geometric point of Y, we have

$$(R^q f_* \mathcal{F})_{\overline{y}} = \varinjlim H^q(V \times_Y X, \mathcal{F}),$$

inductive limit taken over the étale neighborhoods V of $\overline{y}$.

Let $\mathcal{O}_{Y,\overline{y}}$ be the strict localization of Y at $\overline{y}$, $\tilde{Y} = \mathrm{Spec}(\mathcal{O}_{Y,\overline{y}})$ and $\tilde{X} = X \times_Y \tilde{Y}$. We can extend $\mathcal{F}$ to $\tilde{X}$ (this is a particular case of the general notion of an inverse image) in the following manner: let $\tilde{U}$ be an étale scheme on $\tilde{X}$, then there exists an étale neighborhood V of $\overline{y}$ and an étale scheme U over $X \times_Y V$ such that $\tilde{U} = U \times_V \tilde{Y}$; we write

$$\mathcal{F}(\tilde{U}) = \varinjlim F(U \times_V V'),$$

the inductive limit is taken over the étale neighborhoods V' of $\overline{y}$ that contain V. With this definition, we have

$$(R^q f_* F)_{\overline{y}} = H^q(\tilde{X}, \mathcal{F}).$$

The functor f_* has a left adjoint f^*, the "inverse image" functor. If $\overline{x}$ is a geometric point of X and $f(\overline{x})$ its image in Y, we have $(f^*\mathcal{F})_{\overline{x}} = \mathcal{F}_{f(\overline{x})}$. This formula shows that f^* is as an exact functor. The functor f_* thus sends injective sheaves to injective sheaves, and the spectral sequence of the composite functor $\Gamma \circ f_*$ (resp. $g_* f_*$) furnishes the

Theorem 2.3.4 (Leray spectral sequence)**.** *Let $\mathcal{F}$ be an abelian sheaf on $X_{\text{ét}}$ and $f\colon X \longrightarrow Y$ a morphism of schemes (resp. morphisms of schemes $X \xrightarrow{f} Y \xrightarrow{g} Z$). We have a spectral sequence*

$$E_2^{p,q} = H^p(Y, R^q f_* F) \implies H^{p+q}(X, F)$$

$$(\textit{resp.}\ \ E_2^{p,q} = R^p g_* R^q f_* F \implies R^{p+q}(gf)_* F).$$

Corollary 2.3.5. *If $R^q f_* F = 0$ for every $q > 0$, we have $H^p(Y, f_*\mathcal{F}) = H^p(X, \mathcal{F})$ (resp. $R^p g_*(f_*\mathcal{F}) = R^p(gf)_* F$) for every $p \geqslant 0$.*

This applies particularly in the following case:

Proposition 2.3.6. *Let $f\colon X \longrightarrow Y$ be a finite morphism (or even, by passing to the limit, an integral morphism) and $\mathcal{F}$ an abelian sheaf on X. Then $R^q f_* \mathcal{F} = 0$, for every $q > 0$.*

In effect, let $\overline{y}$ be a geometric point of Y, $\tilde{Y}$ the spectrum of the strict localization of Y at y and $\tilde{X} = X \times_Y \tilde{Y}$; by what precedes, it suffices to show that $H^q(\tilde{X}, \mathcal{F}) = 0$ for every $q > 0$. However $\tilde{X}$ is the spectrum of a product of strictly henselian local rings [see [Ray70], chapter I], the functor $\Gamma(\tilde{X}, \bullet)$ is exact because every surjective étale $\tilde{X}$-scheme admits a section, from which follows the assertion.

2.4 Galois cohomology

For $X = \mathrm{Spec}(K)$ the spectrum of a field, we will see that the étale cohomology is identified with the Galois cohomology.

2.4.1. We will start with a topological analogue. If K is the field of functions of an affine integral algebraic variety $Y = \mathrm{Spec}(A)$ over $\mathbb{C}$, we have $K = \varinjlim_{f \in A} A[1/f]$. Said differently $X = \varprojlim U$, U ranging over the set of open subsets of Y. We know that there exist arbitrarily small Zariski open subsets which for the classical topology are $K(\pi, 1)$. We will therefore not be surprised if we consider $\mathrm{Spec}(K)$ itself as a $K(\pi, 1)$, π being the fundamental group (in the algebraic sense) of X, said differently the Galois group of $\overline{K}/K$, where $\overline{K}$ is a separable closure of K.

2.4.2. More precisely, let K be a field, $\overline{K}$ be a separable closure of K and $G = \mathrm{Gal}(\overline{K}/K)$ the topological Galois group. To every finite étale K-algebra A (a finite product of separable extensions of K), we will associate the finite set $\mathrm{Hom}_K(A, \overline{K})$. The Galois group G acts on this set through a discrete quotient (so finite). If $A = K[T]/(F)$, it is identified with the set of roots in $\overline{K}$ of the polynomial F. Galois theory, under the form given to it by Grothendieck, says that:

Proposition 2.4.3. *The functor:*

$$(\textit{finite étale } K\textit{-algebras}) \longrightarrow (\textit{finite sets on which } G \textit{ acts continuously})$$

associating to an étale algebra A, the set $\mathrm{Hom}_K(A, \overline{K})$ is an anti-equivalence of categories.

We deduce from this an analogous description of sheaves for the étale topology on $\mathrm{Spec}(K)$:

Proposition 2.4.4. *The functor:*

$$(\textit{étale sheaves on } \mathrm{Spec}(K)) \longrightarrow (\textit{sets on which } G \textit{ acts continuously})$$

that to a sheaf $\mathcal{F}$ associates its fiber $\mathcal{F}_{\overline{K}}$ at the geometric point $\mathrm{Spec}(\overline{K})$ is an equivalence of categories.

We say that G acts continuously on a set E if the stabilizer of each element of E is an open subgroup of G. The functor in the inverse sense is written in hte evident manner: let A be a finite étale K-algebra, $U = \mathrm{Spec}(A)$ and $U(\overline{K}) = \mathrm{Hom}_K(A, \overline{K})$ the G-set corresponding to A; so we have $\mathcal{F}(U) = \mathrm{Hom}_{G-\mathrm{set}}(U(\overline{K}, \mathcal{F}_{\overline{k}})$.

In particular, if $X = \mathrm{Spec}(K)$, we have $\mathcal{F}(X) = \mathcal{F}_{\overline{K}}^{G}$. If we restrict ourselves to abelian sheaves, we obtain on passing to derived functors, canonical isomorphisms

$$H^q(X_{\text{ét}}, \mathcal{F}) = H^q(G, \mathcal{F}_{\overline{K}}).$$

2.4.5. a) To the constant sheaf $\mathbb{Z}/n$ corresponds $\mathbb{Z}/n$ with trivial G-action.
b) To the sheaf of n-th roots of unity μ_n corresponds the group $\mu_n(\overline{k})$ of n-th roots of unity in $\overline{K}$, with the natural G-action.
c) To the sheaf $\mathbb{G}_m$ corresponds the group $\overline{K}^*$ with the natural G-action.

3 Cohomology of curves

In the case of topological spaces, dévissages using the Künneth formula and simplicial decompositions permit us to come back to calculate the cohomology of the interval $I = [0, 1]$ for which we have $H^0(I, \mathbb{Z}) = \mathbb{Z}$ and $H^q(I, \mathbb{Z}) = 0$ for $q > 0$.

In our case, the dévissages will result in more complicated objects, namely curves on an algebraically closed field; we will calculate their cohomology in this chapter. The situation is more complex than in the topological case since the cohomology groups are trivial only for $q > 2$. The essential ingredient of the calculations is the triviality of the Brauer group of the field of functions on such a curve (Tsen's theorem, 3.2).

3.1 Brauer group

Let us first recall the classical definition:

Definition 3.1.1. Let K be a field and A a K-algebra of finite dimension. We say that A is a central simple algebra on K if the following equivalent conditions are satisfied:

a) A has no nontrivial bilateral ideal and its center is K.

b) There exists a finite Galois extension K'/K such that $A_{K'} = A \otimes_K K'$ is isomorphic to an algebra of square matrices on K'.

c) A is K-isomorphic to an algebra of square matrices on a division ring with center K.

Two such algebras are said to be equivalent if the division rings that are associated to them by c) are K-isomorphic. If these algebras are of the same dimension, this amounts to saying that they are K-isomorphic. The tensor product defines by passing to the quotient, an abelian group structure on the set of equivalence classes. This is the group that we classically call the *Brauer group* of K and we denote it by $\mathrm{Br}(K)$.

3.1.2. We will denote by $\mathrm{Br}(n, K)$ the set of K-isomorphism classes of K-algebras A such that there exists a finite Galois extension K' of K for which A_K is isomorphic to the algebra $M_n(K')$ of $n \times n$ square matrices on K'. By definition $\mathrm{Br}(K)$ is the union of subsets $\mathrm{Br}(n, K)$ for $n \in \mathbb{N}$. Let $\overline{K}$ be an algebraic closure of K and $G = \mathrm{Gal}(\overline{K}/K)$. The set $\mathrm{Br}(n, K)$ is the set of "forms" of $M_n(\overline{K})$, it is therefore canonically isomorphic to $H^1(G, \mathrm{Aut}(M_n(\tilde{K})))$.

We know that every automorphism of $M_n(\overline{K})$ is interior. Consequently the group $\mathrm{Aut}(M_n(\overline{K}))$ is identified with the linear projective group $\mathrm{PGL}(n, \overline{K})$ and we have a canonical bijection:

$$\theta_n : \mathrm{Br}(n, K) \xrightarrow{\sim} H^1(G, \mathrm{PGL}(n, \overline{K}).$$

On the other hand, the exact sequence:

$$1 \longrightarrow \overline{K}^* \longrightarrow \mathrm{GL}(n, \overline{K}) \longrightarrow \mathrm{PGL}(n, \overline{K}) \longrightarrow 1, \qquad (*)$$

allows us to define a boundary operator

$$\Delta_n : H^1(G, \mathrm{PGL}(n, \overline{K})) \longrightarrow H^2(G, \overline{K}^*).$$

Composition θ_n and Δ_n, we obtain a map:

$$\delta_n : \mathrm{Br}(n, K) \longrightarrow H^2(G, \overline{K}^*).$$

We easily verify that the maps δ_n are compatible with each other and define a group homomorphism:

$$\delta : \mathrm{Br}(K) \longrightarrow H^2(G, \overline{K}^*).$$

Proposition 3.1.3. *The homomorphism* $\delta : \mathrm{Br}(K) \longrightarrow H^2(G, \overline{K}^*)$ *is bijective.*

This results in the following two lemmas:

Lemma 3.1.4. *The map* $\Delta_n : H^1(G, \mathrm{PGL}(n, \overline{K})) \longrightarrow H^2(G, \overline{K}^*)$ *is injective.*

By [Ser65], Cor. to Prop. I-44, it suffices to check that every time we twist the exact sequence (*) by an element of $H^1(G, \mathrm{PGL}(n, \overline{K}))$, the H^1 of the middle group is trivial. This middle group is the group of $\overline{K}$-points of the multiplicative group of a simple central algebra A of rank n^2 over K. To prove that $H^1(G, A^*_{\overline{K}})$, we interpret A^* as the group of automorphisms of the free A-module L of rank 1, and H^1 as the set of "forms" of L – of A-modules of rank n^2 over K, automatically free.

Lemma 3.1.5. *Let* $\alpha \in H^2(G, \overline{K}^*)$, K' *a finite extension of* K *contained in* $\overline{K}$, $n = [K' : K]$, *and* $G' = \mathrm{Gal}(\overline{K}/K')$. *If the image of* α *in* $H^2(G', \overline{K}^*)$ *is zero, then,* α *is in the image of* Δ_n.

Note first of all that we have:

$$H^2(G', \overline{K}^*) \simeq H^2(G, (\overline{K} \otimes_K K')^*).$$

[From a geometric point of view if we write $X = \mathrm{Spec}(K)$, $X' = \mathrm{Spec}(K')$ and $\pi : X' \longrightarrow X$ the canonical morphism, we have $R^q\pi_*(\mathbb{G}_{m,X}) = 0$ for $q > 0$ and consequently $H^q(X', \mathbb{G}_{m,X'}) \simeq H^q(X, \pi_*\mathbb{G}_{m,X'}$ for $q \geqslant 0$].

Moreover choosing a base of K' as a vector space over K allows us to define a homomorphism

$$(\overline{K} \otimes_K K')^* \longrightarrow \mathrm{GL}(n, \overline{K})$$

which, to an element x, assigns the endomorphism of multiplication by x of $\overline{K} \otimes_K K'$. We thus have a commutative diagram with rows exact:

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \overline{K}^* & \longrightarrow & (\overline{K} \otimes_K K')^* & \longrightarrow & (\overline{K} \otimes_K K')^*/\overline{K}^* & \longrightarrow & 1 \\ & & \| & & \downarrow & & \downarrow & & \\ 1 & \longrightarrow & \overline{K}^* & \longrightarrow & \mathrm{GL}(n, \overline{K}) & \longrightarrow & \mathrm{PGL}(n, \overline{K}) & \longrightarrow & 1 \end{array}$$

The lemma results in the following commutative diagram that we deduce from above by passing to cohomology:

$$\begin{array}{ccccc} H^1(G, (\overline{K} \otimes_K K')^*/\overline{K}^*) & \longrightarrow & H^2(G, \overline{K}^*) & \longrightarrow & H^2(G, (\overline{K} \otimes_K K')^*) \\ \downarrow & & \| & & \\ H^1(G, \mathrm{PGL}(n, \overline{K})) & \xrightarrow{\Delta_n} & H^2(G, \overline{K}^*) & & \end{array}$$

The knowledge of the Brauer group, particularly its triviality, is extremely important in Galois cohomology as the following proposition shows.

Proposition 3.1.6. *Let K be a field, $\overline{K}$ an algebraic closure of K and $G = \mathrm{Gal}(\overline{K}/K)$. Suppose that, for every finite extension K' of K, we have $\mathrm{Br}(K') = 0$. Then we have:*

(*i*) $H^q(G, \overline{K}^*) = 0$ *for every* $q > 0$.

(*ii*) $H^q(G, F) = 0$ *for every torsion* G*-module* F *and for every* $q \geqslant 2$.

[For the proof, see [Ser62] or [Ser65]].

3.2 Tsen's theorem

Definition 3.2.1. We say that a field K is C_1 if every nonconstant homogenous polynomial $f(x_1, \cdots, x_n)$ of degree $d < n$ has a nontrivial zero.

Proposition 3.2.2. *If a field K is C_1, we have $\mathrm{Br}(K) = 0$.*

It amounts to showing that every division ring D with center K is finite over K and equal to K. Let r^2 be the degree of D over K and $\mathrm{Nrd}\colon D \longrightarrow K$ be the reduced norm. [locally for the étale topology on K, D is isomorphic - noncanonically - to an algebra of matrices M_r and the reduced norm coincides with the determinant map. This is well-defined, independent of the chosen isomorphism between D and M_r because every automorphism of M_r is interior and two similar matrices have the same determinant. This locally

defined map for the étale topology descends, because of its local uniqueness, to a map $\mathrm{Nrd}\colon D \longrightarrow K]$.

The only zero of Nrd is the zero element of D, because, if $x \neq 0$, we have $\mathrm{Nrd}(x)\cdot\mathrm{Nrd}(x^{-1}) = 1$. On the other hand, if $\{e_1, \cdots, e_{r^2}\}$ is a basis of D over K and if $x = \sum_i x_i e_i$, the function $\mathrm{Nrd}(x)$ is written as a homogenous polynomial $\mathrm{Nrd}(x_1, \cdots, x_{r^2})$ of degree r [this is clear locally for the étale topology]. Since K is C_1, we have $r^2 \leqslant r$, that is $r = 1$ and $D = K$.

Theorem 3.2.3 (Tsen)**.** *Let* k *be an algebraically closed field and* K *an extension of transcendance degree* 1 *of* k*. Then* K *is* C_1*.*

Suppose first that $K = k(X)$. Then

$$f(T) = \sum a_{i_1,\cdots,i_n} T_1^{i_1} \cdots T_n^{i_n}$$

a homogenous polynomial of degree $d < n$ with coefficients in $k(X)$. By multiplying the coefficients by a common denominator, we can suppose that they are in $k[X]$. So let $\delta = \sup \deg(a_{i_1\cdots i_n})$. We search for a nontrivial zero in $k[X]$ by the method of undetermined coefficients by writing each T_i $(i = 1, \cdots, n)$ as a polynomial of degree N in X. So the equation $f(T) = 0$ becomes a system of homogenous equations in the $n \times (N+1)$ coefficients of polynomials $T_i(X)$ expressing the nullity of the coefficients of the polynomial in X obtained by replacing T_i by $T_i(X)$. This polynomial is of degree $\delta + Nd$ at most, so there are $\delta + Nd + 1$ equations in $n \times (N+1)$ variables. As k is algebraically closed, this system has a nontrivial solution if $n(N+1) > Nd + \delta + 1$, which is the case for large enough N if $d < n$.

It is clear that, to show the theorem in the general case, it suffices to show it when K is a finite extension of a pure transcendental extension $k(X)$ of k. Let $f(T) = f(T_1, \cdots, T_n)$ be a homogenous polynomial of degree $d < n$ with coefficients in K. Let $s = [K\colon k(X)]$ and $e_1, \cdots, e_s$ a basis for K over $k(X)$. Let us introduce new variables U_{ij}, sn many, such that $T_i = \sum U_{ij} e_j$. For the polynomial $f(T)$ to have a nontrivial zero in K, it suffices that the polynomial $g(X_{ij}) = N_{K/k}(f(T))$ have a nontrivial zero in $k(X)$. However g is a homogenous polynomial of degree sd in sn variables, and the result follows.

Corollary 3.2.4. *Let* k *an algebraically closed field and* K *a transcendence degree* 1 *extension of* k*. Then the étale cohomology groups* $H^q(\mathrm{Spec}(K), \mathbb{G}_m)$ *are trivial for every* $q > 0$*.*

3.3 Cohomology of smooth curves

From now on, without explicit mention of the contrary, the cohomology groups considered are the étale cohomology groups.

Proposition 3.3.1. *Let k be an algebraically closed field and X a connected projective nonsingular curve over k. Then we have:*

$$H^0(X, \mathbb{G}_m) = k^*,$$

$$H^1(X, \mathbb{G}_m) = \mathrm{Pic}(X)$$

,

$$H^q(X, \mathbb{G}_m) = 0 \text{ for } q \geqslant 2.$$

Let η be the generic point of X, $j\colon \eta \longrightarrow X$ be the canonical morphism and $\mathbb{G}_{m,\eta}$ the multiplicative group of the field of fractions $K(X)$ of X. For every closed point x of X, let $i_x\colon x \longrightarrow X$ be the canonical immersion and $\mathbb{Z}_x$ the constant sheaf with value $\mathbb{Z}$ on x. So $j_*\mathbb{G}_{m,\eta}$ is the sheaf of nonzero meromorphic functions on X and $\bigoplus_{x\in X} i_{x*}\mathbb{Z}_x$ the sheaf of divisors, we thus have an exact sequence of sheaves:

$$0 \longrightarrow \mathbb{G}_m \longrightarrow j_*\mathbb{G}_{m,\eta} \xrightarrow{\mathrm{div}} \bigoplus_{x\in X} i_{x*}\mathbb{Z}_x \longrightarrow 0. \qquad (*)$$

Lemma 3.3.2. *We have $R^q j_*\mathbb{G}_{m,\eta} = 0$ for every $q > 0$.*

It suffices to show that the fiber of this sheaf at every closed point x of X is trivial. If $\tilde{\mathcal{O}}_{X,x}$ is the henselization of X at x and K the field of fractions of $\tilde{\mathcal{O}}_{X,x}$, we have

$$\mathrm{Spec}(K) = \eta \times_X \mathrm{Spec}(\tilde{\mathcal{O}}_{X,x}),$$

so $(R^q j_*\mathbb{G}_{m,\eta})_x = H^q(\mathrm{Spec}(K), \mathbb{G}_m)$. However K is an algebraic extension of $k(X)$, so an extension of transcendence degree 1 of k : the lemma results from Cor. 3.2.4.

Lemma 3.3.3. *We have $H^q(X, j_*\mathbb{G}_{m,\eta}) = 0$ for every $q > 0$.*

Indeed from Lem. 3.3.2 and from the Leray spectral sequence for j, we deduce:

$$H^q(X, j_*\mathbb{G}_{m,\eta}) = H^q(\eta, \mathbb{G}_{m,\eta})$$

for every $q \geqslant 0$ and the right hand side is trivial for $q > 0$ by Cor. 3.2.4.

Lemma 3.3.4. *We have $H^q\left(X, \bigoplus_{x\in X} i_{x*}\mathbb{Z}_x\right) = 0$ for every $q > 0$.*

Indeed for every closed point x of X, we have $R^q i_{x*}\mathbb{Z}_x = 0$ for $q > 0$, because i_x is a finite morphism (see Prop. 2.3.6), and

$$H^q(X, i_{x*}\mathbb{Z}_x) = H^q(x, \mathbb{Z}_x).$$

The right hand side is trivial for very $q > 0$, because x is the spectrum of an algebraically closed field [We see that the lemma is true more generally for every "skyscraper" sheaf on X].

We deduce from the preceding lemmas and the exact sequence (*) the equalities:

$$H^q(X, \mathbb{G}_m) = 0 \text{ for } q \geqslant 2,$$

and an exact sequence of cohomology in low degrees:

$$1 \longrightarrow H^0(X, \mathbb{G}_m) \longrightarrow H^0(X, j_*\mathbb{G}_{m,\eta}) \longrightarrow H^0(X, \bigoplus_{x \in X} i_{x*}\mathbb{Z}_x) \longrightarrow H^1(X, \mathbb{G}_m) \longrightarrow 1$$

which is no different than the exact sequence:

$$1 \longrightarrow k^* \longrightarrow k(X)^* \longrightarrow \mathrm{Div}(X) \longrightarrow \mathrm{Pic}(X) \longrightarrow 1.$$

From Prop. 3.3.1 we deduce that the cohomology groups of X with coefficients in $\mathbb{Z}/n$, n coprime to the characteristic of k, have a reasonable form:

Corollary 3.3.5. *If X is of genus g and if n is invertible in k, $H^q(X, \mathbb{Z}/n) = 0$ for $q > 2$, and free on $\mathbb{Z}/n$ of rank $1, 2g, 1$ for $q = 0, 1, 2$. Replacing $\mathbb{Z}/n$ by the isomorphic group μ_n, we have canonical isomorphisms:*

$$H^0(X, \mu_n) = \mu_n$$
$$H^1(X, \mu_n) = \mathrm{Pic}^0(X)_n$$
$$H^2(X, \mu_n) = \mathbb{Z}/n.$$

As the field k is algebraically closed, $\mathbb{Z}/n$ is isomorphic (noncanonically) to μ_n. From the Kummer exact sequence:

$$0 \longrightarrow \mu_n \longrightarrow \mathbb{G}_m \longrightarrow \mathbb{G}_m \longrightarrow 0,$$

and from Prop. 3.3.1, we deduce the equalities:

$$H^q(X, \mathbb{Z}/n) = 0 \text{ for } q > 2,$$

and, in low degrees, the exact sequences:

$$0 \longrightarrow H^0(X, \mu_n) \longrightarrow k^* \xrightarrow{n} k^* \longrightarrow 0$$

$$0 \longrightarrow H^1(X, \mu_n) \longrightarrow \mathrm{Pic}(X) \xrightarrow{n} \mathrm{Pic}(X) \longrightarrow H^2(X, \mu_n) \longrightarrow 0$$

Moreover we have an exact sequence:

$$0 \longrightarrow \mathrm{Pic}^0(X) \longrightarrow \mathrm{Pic}(X) \xrightarrow{\deg} \mathbb{Z} \longrightarrow 0,$$

and $\mathrm{Pic}^0(X)$ is identified with the group of rational points over k of an abelian variety of dimension g, the Jacobian of X. In such a group, multiplication by n is surjective and its kernel is a free $\mathbb{Z}/n\mathbb{Z}$-module of rank $2g$ (since n is invertible in k); from which follows the corollary.

A clever dévissage, using the "trace method," permits us to obtain the following corollary.

Proposition 3.3.6 ([SGA4V3, IX §5.7]). *Let* k *be an algebraically closed field,* X *an algebraic curve over* k *and* F *a torsion sheaf on* X*. Then:*

(*i*) *We have* $H^q(X, F) = 0$ *for* $q > 2$.

(*ii*) *If* X *is affine, we also have* $H^q(X, F) = 0$ *for* $q > 1$.

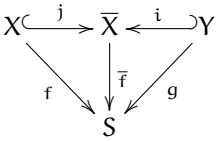
For the proof, and for an exposition on the "trace method," we refer the reader to [SGA4V3, IX §5].

3.4 Dévissages

To calculate the cohomology of varieties of dimension > 1 we employ fibrations by curves, which allows us to study the morphisms whose fibers are of dimension $\leqslant 1$. This principle possesses several variants, let us indicate some of them.

3.4.1. Let A be k-algebra of finite type and $a_1, \cdots, a_n$ generators of A. If we write $X_0 = \mathrm{Spec}(k)$, $X_i = \mathrm{Spec}(k[a_1, \cdots, a_i])$, $X_n = \mathrm{Spec}(A)$, the canonical inclusions $k[a_1, \cdots, a_i] \longrightarrow k[a_1, \cdots, a_i, a_{i+1}]$ define morphisms $X_n \longrightarrow X_{n-1} \longrightarrow \cdots \longrightarrow X_1 \longrightarrow X_0$ whose fibers are of dimension $\leqslant 1$.

3.4.2. In the case of a smooth morphism, we can be more precise. We mean by an *elementary fibration* a morphism of schemes $f\colon X \longrightarrow S$ which can be placed in a commutative diagram:



satisfying the following conditions:

(i) j is a dense open immersion into each fiber and $X = \overline{X} - Y$.

(ii) $\overline{f}$ is projective and smooth, with fibers irreducible, geometric, and of dimension 1.

(iii) g is an étale covering and no fiber of g is empty.

We mean by a *good neighborhood* relative to S an S-scheme X such that there exist S-schemes $X = X_n, \cdots, X_0 = S$ and elementary fibrations $f_i\colon X_i \longrightarrow X_{i-1}$, $i = 1, \cdots, n$. We can show [SGA4V3, XI §3.3] that if X is a smooth scheme on an algebraically closed k every rational point of X has an open neighborhood that is a good neighborhood (relative to $\mathrm{Spec}(k)$).

3.4.3. We can dévisse a proper morphism $f\colon X \longrightarrow S$ in the following manner. By Chow's lemma, there exists a commutative diagram

$$\begin{array}{ccc} X & \xleftarrow{\pi} & \overline{X} \\ & {\scriptstyle f}\searrow \quad \swarrow{\scriptstyle \overline{f}} & \\ & S & \end{array}$$

where π and $\overline{f}$ are projective morphisms, π is moreover an isomorphism over a dense open subset of X. Locally on S, $\overline{X}$ is a closed subscheme of a projective space of the type $\mathbb{P}^n_S$.

We dévisse the latter by considering the projection $\varphi\colon \mathbb{P}^n_S \longrightarrow \mathbb{P}^1_S$ which sends the point with homogenous coordinates $(x_0, x_1, \cdots, x_n)$ on (x_0, x_1). This is a rational map defined outside the closed subset $Y \simeq \mathbb{P}^{n-2}_S$ of $\mathbb{P}^n_S$ defined by the homogenous equations $x_0 = x_1 = 0$. Let $u\colon P \longrightarrow \mathbb{P}^n_S$ be the blow-up at Y; the fibers of u are of dimension $\leqslant 1$. Moreover there exists a natural morphism $v\colon P \longrightarrow \mathbb{P}^1_S$ that extends the rational map φ and v makes P a $\mathbb{P}^1_S$-scheme locally isomorphic to the projective space of type $\mathbb{P}^{n-1}$ which can in turn be projected onto a $\mathbb{P}^1$, etc.

3.4.4. We can sweep a smooth projective variety X by a *Lefschetz pencil*. The blowup $\tilde{X}$ of the intersection of the axis of the pencil with X projects onto $\mathbb{P}^1$ and the fibers of this projection are the hyperplane sections of X by the hyperplanes of the pencil.

4 Base-change theorem for a proper morphism

4.1 Introduction

This section is devoted to the proof and applications of:

Theorem 4.1.1. *Let $f\colon X \longrightarrow S$ be a proper morphism of schemes and $\mathcal{F}$ an abelian torsion sheaf on X. Then, for any $q \geqslant 0$, the fiber of $R^q f_* \mathcal{F}$ at a geometric point s of S is isomorphic to the cohomology $H^q(X_s, \mathcal{F})$ of the fiber $X_s = X \otimes_S \operatorname{Spec} k(s)$ of f at s.*

For $f\colon X \longrightarrow S$ a proper and separated continuous map (separated meaning that the diagonal of $X \times_S X$ is closed) between topological spaces, and F an abelian sheaf on X, the analogous result is well-known, and elementary: as f is closed, the $f^{-1}(V)$'s for neighborhoods V of s form a fundamental system of neighborhoods of X_s, and we verify that $H^*(X_s, \mathcal{F}) = \varinjlim_U H^*(U, \mathcal{F})$, for U ranging over the neighborhoods of X_s. In practice, X_s also has a fundamental system $\mathcal{U}$ of neighborhoods U from which it is deformation retracted and, for F constant, we thus have $H^*(X_s, \mathcal{F}) = H^*(U, \mathcal{F})$. In figurative terms: the special fiber swallows the general fiber.

In the case of schemes the proof is more delicate and it is indispensable to suppose that $\mathcal{F}$ is of torsion ([SGA4V3, XII §2]). Considering the description of the fibers of $R^q f_* \mathcal{F}$ (see 2.3.3.), Thm. 4.1.1 is essentially equivalent to

Theorem 4.1.2. *Let A be a strictly henselian local ring and $S = \operatorname{Spec}(A)$. Let $f\colon X \longrightarrow S$ a proper morphism and X_0 the closed fiber of f. Then, for every abelian torsion sheaf $\mathcal{F}$ on X and for every $q \geqslant 0$, we have $H^q(X, \mathcal{F}) \xrightarrow{\sim} H^q(X_0, \mathcal{F})$.*

By passing to the limit we see that it suffices to show the theorem when A is the strict henselization of a $\mathbb{Z}$-algebra of finite type at a prime ideal. We first treat the case $q = 0$ or 1 and $\mathcal{F} = \mathbb{Z}/n$ (see 4.2). An argument based on the notion of constructible sheaves (see 4.3) shows that it actually suffices to consider the case where $\mathcal{F}$ is constant. On the other hand the dévissage (see 3.4.3.) allows us to suppose that X_0 is a curve; in this case it only remains to show the theorem for $q = 2$ (see 4.4).

Among other applications (see 4.6), the theorem allows us to define the notion of cohomology with proper support (see 4.5).

4.2 Proof for $q = 0$ or 1 and $\mathcal{F} = \mathbb{Z}/n$

The result for $q = 0$ and $\mathcal{F}$ constant is equivalent to the following proposition [the Zariski connection theorem]:

Proposition 4.2.1. *Let A be a noetherian and henselian local ring and $S = \operatorname{Spec}(A)$. Let $f\colon X \longrightarrow S$ be a proper morphism and X_0 the closed fiber of f. Then the sets of connected components $\pi_0(X)$ and $\pi_0(X_0)$ are in bijection.*

It boils down to the same thing to show that the sets of clopen subspaces $\mathrm{Clo}(X)$ and $\mathrm{Clo}(X_0)$ are in bijection. We know that the set $\mathrm{Clo}(X)$ corresponds bijectively to the set of idempotents of $\Gamma(X, \mathcal{O}_X)$, similarly $\mathrm{Clo}(X_0)$ corresponds bijectively to the set of idempotents of $\Gamma(X_0, \mathcal{O}_{X_0})$. It is therefore a question of showing that the canonical map

$$\mathrm{Idem}\Gamma(X, \mathcal{O}_X) \longrightarrow \mathrm{Idem}\Gamma(X_0, \mathcal{O}_{X_0})$$

is bijective.

We will denote by $\mathfrak{m}$ the maximal ideal of A, $\Gamma(X, \mathcal{O}_X)^\wedge$ the completion of $\Gamma(X, \mathcal{O}_X)$ for the $\mathfrak{m}$-adic topology and, for every integer $n \geqslant 0$, $X_n = X \otimes_A A/\mathfrak{m}^{n+1}$. By the theorem of finiteness for proper morphisms (see [EGA3], 3.2), $\Gamma(X, \mathcal{O}_X)$ is a finite A-algebra; since A is henselian, it follows that the canonical map

$$\mathrm{Idem}\Gamma(X, \mathcal{O}_X) \longrightarrow \mathrm{Idem}\Gamma(X, \mathcal{O}_X)^\wedge$$

is bijective.

By the comparison theorem for proper morphisms (see [EGA3] 4.1), the canonical map

$$\Gamma(X, \mathcal{O}_X)^\wedge \longrightarrow \varprojlim \Gamma(X_n, \mathcal{O}_{X_n})$$

is bijective. In particular the canonical map

$$\mathrm{Idem}\Gamma(X, \mathcal{O}_X)^\wedge \longrightarrow \varprojlim \mathrm{Idem}\Gamma(X_n, \mathcal{O}_{X_n})$$

is bijective. But, since X_n and X_0 have the same underlying topological space, the canonical map

$$\mathrm{Idem}\Gamma(X_n, \mathcal{O}_{X_n}) \longrightarrow \mathrm{Idem}\Gamma(X_0, \mathcal{O}_{X_0})$$

is bijective for every n, which completes the proof.

Since $H^1(X, \mathbb{Z}/n)$ is in bijection with the set of isomorphism classes of galois étale coverings of X of group $\mathbb{Z}/n$, the theorem for $q = 1$ and $\mathcal{F} = \mathbb{Z}/n$ results in the following proposition.

Proposition 4.2.2. *Let A be a noetherian and henselian local ring and $S = \mathrm{Spec}(A)$. Let $f\colon X \longrightarrow S$ be a proper morphism and X_0 the closed fiber of f. Then the restriction functor*

$$\mathrm{Et.cov.}(X) \longrightarrow \mathrm{Et.cov.}(X_0)$$

is an equivalence of categories.

[If X_0 is connected and if we choose a geometric point of X_0 as the basepoint, that is to say that the canonical map $\pi_1(X_0) \longrightarrow \pi_1(X)$ on the (profinite) fundamental groups is bijective].

Prop. 4.2.1 shows that this functor is fully faithful. Indeed, if X' and X'' are two étale covers of X, an X-morphism from X' to X'' is determined by its graph which is a clopen subspace of $X \times_X X''$.

It is therefore a question of showing that every étale cover X'_0 of X_0 extends to an étale cover of X. We know that the étale covers are independent of nilpotent elements (see [SGA1, Ch. I]), consequently X'_0 lifts uniquely to an étale cover X'_n of X_n for every $n \geqslant 0$, said otherwise to an étale cover $\mathcal{X}'$ of the formal scheme $\mathcal{X}$ the completion of X along X_0. By Grothendieck's theorem of algebrization of formal coherent sheaves (see the existence theorem of [EGA3] Ch. 5), $\mathcal{X}'$ is the formal completion of an étale cover $\overline{X}'$ of $\overline{X} = X \otimes_A \hat{A}$.

By passing to the limit, it suffices to show the proposition in the case where A is the henselization of a $\mathbb{Z}$-algebra of finite type. We can then apply Artin's approximation theorem to the functor $F\colon \mathbf{Alg}_{\hat{A}} \longrightarrow \mathbf{Set}$ which, to an A-algebra B, assigns the set of isomorphism classes of étale covers of $X \otimes_A B$. Indeed this functor is locally of finite presentation: if B_i is an inductive filtered system of A-algebras and if $B = \varinjlim B_i$, we have $F(B) = \varinjlim F(B_i)$. By Artin's theorem, given an element $\overline{\xi} \in F(\hat{A})$, in this case the isomorphism class of $\overline{X}'$, there exists $\xi \in F(A)$ having the same image of $\overline{\xi}$ in $F(A/\mathfrak{m})$. Said otherwise there exists an étale cover X' of X whose restriction to X_0 is isomorphic to X'_0.

4.3 Constructible sheaves

In this paragraph, we consider a *noetherian* scheme X and we mean by a sheaf on X an *abelian* sheaf on X_{et}.

Definition 4.3.1. We say that a sheaf $\mathcal{F}$ on X is *locally constant constructible* (in short, l.c.c.) if it is represented by an étale cover of X.

Definition 4.3.2. We say that a sheaf $\mathcal{F}$ on X is *constructible* if it satisfies the following equivalent conditions:

(i) There exists a finite surjective family of subschemes X_i of X such that the restriction of $\mathcal{F}$ to X_i is l.c.c..

(ii) There exists a finite family of finite morphisms $p_i\colon X'_i \longrightarrow X$, for eachc i a constant constructible sheaf (= defined by a finite abelian group) $\mathcal{C}_i$ on X'_i, and a monomorphism $\mathcal{F} \longrightarrow \prod_i p_{i*}\mathcal{C}_i$.

We easily check that the category of constructible sheaves on X is an abelian category. Moreover, if $u\colon \mathcal{F} \longrightarrow \mathcal{G}$ is a homomorphism of sheaves and if $\mathcal{F}$ is constructible, the sheaf $\mathrm{Im}(u)$ is constructible.

Lemma 4.3.3. *Every torsion sheaf $\mathcal{F}$ is an inductive filtered limit of constructible sheaves.*

Indeed, if $j\colon U \longrightarrow X$ is an étale scheme of finite type on X, an element $\xi \in \mathcal{F}(U)$ such that $n\xi = 0$ defines a homomorphism of sheaves $j_!\underline{\mathbb{Z}/n}_U \longrightarrow \mathcal{F}$ whose image (the smalles subsheaf of $\mathcal{F}$ whose ξ is a local section) is a constructible subsheaf of $\mathcal{F}$. It is clear that $\mathcal{F}$ is an inductive limit of such subsheaves.

Definition 4.3.4. Let $\mathcal{C}$ be an abelian category and T a functor defined on $\mathcal{C}$ taking values in the category of abelian groups. We will say that T is *effaceable* in $\mathcal{C}$ if, for every object A of $\mathcal{C}$ and every $\alpha \in T(A)$, there exists a monomorphism $u\colon A \longrightarrow M$ in $\mathcal{C}$ such that $T(u)\cdot\alpha = 0$.

Lemma 4.3.5. *The functors $H^q(X,\cdot)$ for $q > 0$ are effaceable in the category of constructible sheaves on X.*

It suffices to remark that, if $\mathcal{F}$ is a constructible sheaf, there necessarily exists an integer $n > 0$ such that $\mathcal{F}$ is a sheaf of $\mathbb{Z}/n$-modules. So there exists a monomorphism $\mathcal{F} \hookrightarrow \mathcal{G}$, where $\mathcal{G}$ is a sheaf of $\mathbb{Z}/n$-modules and $H^q(X,G) = 0$ for every $q > 0$. We can for example take for G to be the Godement resolution $\prod_{x\in X} i_{x*}\mathcal{F}_{\overline{x}}$, where x ranges over the points of X and $i_x\colon \overline{x} \longrightarrow X$ is a geometric point centered at x. By Lem. 4.3.3 $\mathcal{G}$ is an inductive limit of constructible sheaves, from which the lemma follows, since the functors $H^q(X,\cdot)$ commute with inductive limits.

Lemma 4.3.6. *Let $\varphi^\bullet\colon T^\bullet \longrightarrow T'^\bullet$ be a morphism of cohomology functors defined on an abelian category $\mathcal{C}$ and taking values in the category of abelian groups. Suppose that T^q is effaceable for $q > 0$ and let $\mathcal{E}$ be a subset of objects of $\mathcal{C}$ such that every object of $\mathcal{C}$ is contained in an object of $\mathcal{E}$. Then the following conditions are equivalent:*

(i) $\varphi^q(A)$ *is bijective for every* $q \geqslant 0$ *and every* $A \in \mathrm{Ob}\mathcal{C}$.

(ii) $\varphi^0(M)$ *is bijective and* $\varphi^q(M)$ *surjective for every* $q > 0$ *and every* $M \in \mathcal{E}$.

(iii) $\varphi^0(A)$ *is bijective for every* $A \in \mathrm{Ob}\mathcal{C}$ *and* $(T')^q$ *is effaceable for every* $q > 0$.

The proof is by recurrence on q and does not present difficulties.

Proposition 4.3.7. *Let X_0 be a subscheme of X. Suppose that, for every $n \geqslant 0$ and for every scheme X' finite over X, the canonical map*

$$H^q(X', \mathbb{Z}/n) \longrightarrow H^q(X'_0, \mathbb{Z}/n),$$

where $X'_0 = X' \times_X X_0$, is bijective for $q = 0$ and surjective for $q > 0$. Then, for every torsion sheaf $\mathcal{F}$ on X and for every $q \geqslant 0$, the canonical map

$$H^q(X, \mathcal{F}) \longrightarrow H^q(X_0, \mathcal{F})$$

is bijective.

By passing to the limit, it suffices to the assertion for $\mathcal{F}$ constructible. We apply Lem. 4.3.6 by taking $\mathcal{C}$ to be the category of constructible sheaves on X, $T^q = H^q(X, \cdot)$, $(T')^q = H^q(X_0, \cdot)$ and $\mathcal{E}$ the set of constructible sheaves of the form $\pi_{p_i^*}\mathcal{C}_i$, where $p_i : X_i' \longrightarrow X$ is a finite morphism and $\mathcal{C}_i$ is a constant sheaf finite on X_i'.

4.4 End of the proof

By the method of fibration by curves (3.4.3.), we come back to proving the theorem in relative dimension $\leqslant 1$. By the preceding paragraph, it will suffice to prove that, if S is the spectrum of a local strictly henselian noetherian ring, $f : X \longrightarrow S$ a proper morphism whose closed fiber X_0 is of dimension $\leqslant 1$ and n an integer $\geqslant 0$, the canonical homomorphism

$$H^q(X, \mathbb{Z}/n) \longrightarrow H^q(X_0, \mathbb{Z}/n)$$

is bijective for $q = 0$ and surjective for $q > 0$.

The cases $q = 0$ and 1 were seen above and we have $H^q(X_0, \mathbb{Z}/n) = 0$ for $q \geqslant 3$; it suffices then to treat the case $q = 2$. Evidently we can suppose that n is a power of a prime number. If $n = p^r$, where p is the characteristic of the residual field of S, the theory of Artin-Schreier shows that we have $H^2(X_0, \mathbb{Z}/p^r) = 0$. If $n = \ell^r$, $\ell \neq p$, we deduce from Kummer's theory a commutative diagram

$$\begin{array}{ccc} \mathrm{Pic}(X) & \xrightarrow{\alpha} & H^2(X, \mathbb{Z}/\ell^r) \\ \downarrow & & \downarrow \\ \mathrm{Pic}(X_0) & \xrightarrow{\beta} & H^2(X_0, \mathbb{Z}/\ell^r) \end{array}$$

where the map β is surjective [We have seen this in section 3 for a smooth curve on an algebraically closed field, but similar arguments apply to any curve on a separably closed field].

To conclude, it suffices to show:

Proposition 4.4.1. *Let S be the spectrum of a henselian noetherian local ring and $f : X \longrightarrow S$ a proper morphism whose closed fiber X_0 is of dimension $\leqslant 1$. Then the canonical restriction map*

$$\mathrm{Pic}(X) \longrightarrow \mathrm{Pic}(X_0)$$

is surjective [*It is sufficient that the morphism f is* separated *of finite type*].

To simplify the proof, we suppose that X is integral, even though this is not necessary. Every invertible sheaf on X_0 is associated to a Cartier divisor (because X_0 is a curve, hence quasi-projective), it suffices then to show that the canonical map $\mathrm{Div}(X) \longrightarrow \mathrm{Div}(X_0)$ is surjective.

Every divisor on X_0 is a linear combination of divisors whose support is concentrated at a single closed non-isolated point of X_0. Let x be one such point, $t_0 \in \mathcal{O}_{X_0,x}$ be a regular non-invertible element of $\mathcal{O}_{X_0,x}$ and D_0 be the divisor concentrated at x of local equation t_0. Let U be an open neighborhood of x in X such that there exists a section $t \in \Gamma(U, \mathcal{O}_U)$ relevant to t_0. Let Y be the closure of U of equation $t = 0$; even if it means taking U to be small enough, we can suppose that x is the only point of $Y \cap X_0$. Then Y is quasi-finite over S at x; since S is the spectrum of a henselian local ring, we can deduce that $Y = Y_1 \coprod Y_2$, where Y_1 is finite over S and where Y_2 does not meet X_0. Moreover, as X is separated over S, Y_1 is closed in X.

Even if it means replacing U by a smaller open neighborhood of x, we can suppose that $Y = Y_1$, said otherwise, that Y is closed in X. We can then define a divisor D on X associated to D_0 by supposing $D|_{X-Y} = 0$ and $D|_U = \mathrm{div}(t)$ which makes sense because t is invertible on $U - Y$.

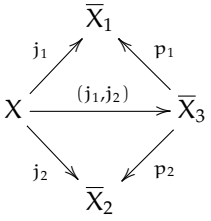
Remark 4.4.2. In the case where f is proper, we can conduct a proof similar to that of Prop. 4.2.2. Indeed, as X_0 is a curve, there exists no obstruction to lift an invertible sheaf on X_0 to infinitesimal neighborhoods X_n of X_0, so to the formal completion $\mathfrak{X}$ of X along X_0. We conclude then by successively applying Grothendieck's existence theorem and Artin's approximation theorem.

4.5 Cohomology with proper support

Definition 4.5.1. Let X be a separated scheme of finite type on a field k. By a theorem of Nagata, there exists a proper scheme $\overline{X}$ over k and an open immersion $j\colon X \longrightarrow \overline{X}$. For every torsion sheaf $\mathcal{F}$ on X we note $j_!\mathcal{F}$ the extension by 0 of $\mathcal{F}$ to $\overline{X}$ and we define the cohomology groups of proper support $H^q_c(X, \mathcal{F})$ by writing

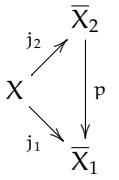
$$H^q_c(X, \mathcal{F}) = H^q(\overline{X}, j_!\mathcal{F}).$$

We will show that this definition is independent of the chosen compactification $j\colon X \longrightarrow \overline{X}$. Let $j_1\colon X \longrightarrow \overline{X}_1$ and $j_2\colon X \longrightarrow \overline{X}_2$ be two compactifications. Then X sends itself into $\overline{X}_1 \times \overline{X}_2$ by $x \mapsto (j_1(x), j_2(x))$ and the closed image $\overline{X}_3$ of X by this map is a compactification of X. We thus have a commutative diagram:



p_1 and p_2, the restrictions of natural projections to $\overline{X}_3$, are proper morphisms.

It suffices then to treat the case where we have a commutative diagram



Lemma 4.5.2. *We have* $p_*(j_{2!}\mathcal{F}) = j_{1!}\mathcal{F}$ *and* $R^q p_*(j_{2!}\mathcal{F}) = 0$, *for* $q > 0$.

We note right away that the lemma suffices to conclude our claim. Using the Leray spectral sequence of the morphism p, we deduce that we have, for every $q \geqslant 0$,

$$H^q(\overline{X}_2, j_{2!}\mathcal{F}) = H^q(\overline{X}_1, j_{1!}\mathcal{F}).$$

To show the lemma, we reason fiber by fiber using the base-change theorem (see 4.1.1) for p. The result is immediate, since, over a point of X, p is an isomorphism at, over a point of $\overline{X}_1 - X$, $j_{2!}\mathcal{F}$ is trivial over the fiber of p. with p a proper morphism.

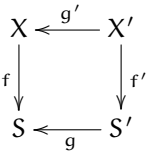
4.5.3. Likewise, if $f\colon X \longrightarrow S$ is a separated morphism of finite type between noetherian schemes, there exists a proper morphism $\overline{f}\colon \overline{X} \longrightarrow S$ and an open immersion $j\colon X \longrightarrow \overline{X}$. We define then the superior direct images with proper support $R^q f_!$ by writing for every torsion sheaf $\mathcal{F}$ on X

$$R^q f_! = R^q f_*(j_!\mathcal{F}).$$

We verify as before that this definition is independent of the chosen compactification.

Theorem 4.5.4. *Let* $f\colon X \longrightarrow S$ *be a separated morphism of finite type between noetherian schemes and* $\mathcal{F}$ *a torsion sheaf on* X. *Then the fiber of* $R^q f_!\mathcal{F}$ *at a geometric point* s *of* S *is isomorphic to the cohomology with proper support* $H^q_c(X_s, \mathcal{F})$ *of the fiber* X_s *of* f *at* s.

This is a simple variant of the base-change theorem for a proper morphism (see Thm. 4.1.1). More generally, if



is a cartesian diagram, we have a canonical isomorphism.

$$g^*(R^q f_!\mathcal{F}) \cong R^q f'_!(g'^*\mathcal{F}). \tag{4.5.4.1}$$

4.6 Applications

Theorem 4.6.1. [*Annihilation theorem*] *Let* $f\colon X \longrightarrow S$ *be a separated morphism of finite type whose fibers are of dimension* $\leqslant n$ *and* $\mathcal{F}$ *a torsion sheaf on* X*. Then we have* $R^q f_!\mathcal{F} = 0$ *for* $q > 2n$.

By the base-change theorem, we can suppose that S is the spectrum of a separably closed field. If $\dim X = n$, there exists an affine open subset U of X such that $\dim(X - U) < n$; we then have an exact sequence $0 \longrightarrow \mathcal{F}_U \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}_{X-U} \longrightarrow 0$ and, by recurrence on n, it suffices to prove the theorem for $X = U$ affine. Then the method of fibration by curves (see 3.4.1.) and the base-change theorem allows us to reduce to a curve on a separably closed field for which we deduce the required result from Tsen's theorem (see Prop. 3.3.6).

Theorem 4.6.2. *Let* $f\colon X \longrightarrow S$ *be a separated morphism of finite type and* $\mathcal{F}$ *a constructible sheaf on* X*. Then the sheaves* $R^q f_!\mathcal{F}$ *are constructibles.*

We will only consider the case where $\mathcal{F}$ is annihilated by an invertible integer over X.

To prove the theorem we reduce to the case where $\mathcal{F}$ is a constant sheaf $\underline{\mathbb{Z}/n}$ and where $f\colon X \longrightarrow S$ is a proper smooth morphism whose fibers are geometrically connected curves of genus g. For n invertible over X, the sheaves $R^q f_*\mathcal{F}$ are then locally free of finite rank, trivial for $q > 2$ (Thm. 4.6.1). Replacing $\mathbb{Z}/n$ by the sheaf locally isomorphic (over S) to μ_n, we have canonically

$$\begin{aligned} R^0 f_*\mu_n &= \mu_n \\ R^1 f_*\mu_n &= \underline{\mathrm{Pic}}(X/S)_n \\ R^2 f_*\mu_n &= \mathbb{Z}/n. \end{aligned} \tag{6.2.1}$$

Theorem 4.6.3 (Comparison with classical cohomology)**.** *Let* $f\colon X \longrightarrow S$ *be a separated morphism between schemes of finite type over* $\mathbb{C}$*, and* $\mathcal{F}$ *a torsion sheaf over* X*. Let us denote by an exponent* ^{an} *the functor of passing to usual topological spaces, and by* $R^q f_!^{an}$ *the derived functors of the direct image functor with proper support by* f^{an}*. We have*

$$(R^q f_!\mathcal{F})^{an} \cong R^q f_!^{an}\mathcal{F}^{an}.$$

In particular, for $S =$ *a point and* $\mathcal{F}$ *the constant sheaf* $\mathbb{Z}/n$*,*

$$H_c^q(X, \mathbb{Z}/n) \cong H_c^q(X^{an}, \mathbb{Z}/n).$$

Dévissages using the base-change theorem allow us to reduce to the case where X is a proper smooth curve, where S = a point, and where $\mathcal{F} = \mathbb{Z}/n$. The cohomology groups considered are then trivial for $q \neq 0, 1, 2$ and we

invoke GAGA: indeed, if X is proper over $\mathbb{C}$, we have $\pi_0(X) = \pi_0(X^{an})$ and $\pi_1(X) =$ profinite completion of $\pi_1(X^{an})$, from which the assertion for $q = 0, 1$. For $q = 2$, we use the Kummer exact sequence and the fact that, by GAGA again, $\mathrm{Pic}(X) = \mathrm{Pic}(X^{an})$.

Theorem 4.6.4 (Cohomological dimension of affine schemes)**.** *Let X be an affine scheme of finite type over a separably closed field and $\mathcal{F}$ a torsion sheaf on X. We then have $H^q(X, \mathcal{F}) = 0$ for $q > \dim(X)$.*

For a very pretty proof we refer the reader to [SGA4V3, XIV §2 and 3].

Remark 4.6.5. This theorem is in some way a substitute for Morse theory. We will indeed consider the classical case where X is smooth and affine over $\mathbb{C}$ immersed in an affine space of type $\mathbb{C}^N$. Then, for almost every point $p \in \mathbb{C}^N$, the function "distance to p" on X is a Morse function and the indices of its critical points are smaller than $\dim(X)$. So X is obtained by reattaching the handles of index smaller than $\dim(X)$, from which the classical analogue of Thm. 4.6.4 follows.

5 Local acyclicity of smooth morphisms

Let X be a complex analytic variety and $f\colon X \longrightarrow D$ a morphism from X to the unit disk. We denote by $[0, t]$ the right closed segment with endpoints 0 and t in D and $]0, t]$ the semi-open segment. If f is *smooth*, the inclusion

$$j\colon f^{-1}(]0, t]) \hookrightarrow f^{-1}([0, t])$$

is a homotopy equivalence; we can push the special fiber $X_0 = f^{-1}(0)$ into $f^{-1}([0, t])$.

In practice, for small enough t, $f^{-1}(]0, t])$ will be fibered over $]0, t]$ such that the inclusion

$$X_t = f^{-1}(t) \hookrightarrow f^{-1}(]0, t])$$

will also be a homotopy equivalence. We then call the following homotopy class of maps the *cospecialization morphism*:

$$\mathrm{cosp}\colon X_0 \hookrightarrow f^{-1}([0, t]) \xleftarrow{\simeq} f^{-1}(]0, t]) \xleftarrow{\simeq} X_t$$

We can express this construction in terms of images by saying that, for a smooth morphism, the general fiber contains the special fiber.

Let us no longer suppose that f is necessarily smooth (but let us suppose that $f^{-1}(]0, t])$ is fibered over $]0, t]$). We can define again a morphism cosp^* in cohomology when $j_*\mathbb{Z} = \mathbb{Z}$ and $R^q j_*\mathbb{Z} = 0$ for $q > 0$. Under these hypotheses, the Leray spectral sequence for j shows that we have

$$H^*(f^{-1}([0, t]), \mathbb{Z}) \xrightarrow{\sim} H^*(f^{-1}(]0, t]), \mathbb{Z})$$

and cosp^* is the composite morphism:

$$\mathrm{cosp}^*\colon H^*(X_t,\mathbb{Z}) \xleftarrow{\sim} H^*(f^{-1}(]0,t]),\mathbb{Z}) \xleftarrow{\sim} H^*(f^{-1}([0,t]),\mathbb{Z}) \longrightarrow H^*(X_0,\mathbb{Z})$$

The fiber of $R^qj_*\mathbb{Z}$ at a point $x \in X_0$ can be calculated as follows. We take a ball B_ε centered at x with small enough radius ε in an ambient space, and for η small enough, we suppose $E = X \cap B_\varepsilon \cap f^{-1}(\eta t)$; this is the *variety of evanescent cycles* at x. We have:

$$(R^qj_*\mathbb{Z})_x \xleftarrow{\sim} H^q(X \cap B_\varepsilon \cap f^{-1}(]0,\eta t]),\mathbb{Z}) \xrightarrow{\sim} H^q(E,\mathbb{Z})$$

and the cospecialization morphism is defined in cohomology as long as the varieties of evanescent cycles are acyclic [$H^0(E,\mathbb{Z}) = \mathbb{Z}$ and $H^q(E,Z) = 0$ for $q > 0$], which can be expressed by saying that f is *locally acyclic*.

This chapter is devoted to the analogue of this situation for a smooth morphism of schemes and for the étale cohomology. However it is essential in this body of work to limit ourselves to torsion coefficients and to *first order residual characteristics*. Paragraph 1 is devoted to generalities on locally acyclic morphisms and cospecialization arrows. In paragraph 2, we demonstrate that a smooth morphism is locally acyclic. In paragraph 3, we reconcile this result with those from the preceding chapter to deduce two applications: a theorem of specialization of cohomology groups (the cohomology of geometric fibers of a proper smooth morphism is locally constant) and a theorem for base-change by a smooth morphism.

In all that follows, we fix an integer n and the word "scheme" will be used to mean "scheme on which n is invertible." "Geometric point" will always mean an "algebraic geometric point" (see 2.3) $x\colon \mathrm{Spec}(k) \longrightarrow X$, with k algebraically closed.

5.1 Locally acyclic morphisms

Notation 5.1.1. Given a scheme S and a geometric point s of S, we denote by $\tilde{S}^s$ the strictly localized spectrum of S at s.

Definition 5.1.2. We say that a geometric point t of S is a generalization of s if it is defined by an algebraic closure of the residual field of a point in $\tilde{S}^s$. We also say that s is a specialization of t and we call the S-morphism $t \longrightarrow \tilde{S}^s$ the specialization arrow.

Definition 5.1.3. Let $f\colon X \longrightarrow S$ be a morphism of schemes. Let s be a geometric point of S, t a generalization of s, x a geometric point of X over s and $\tilde{X}^x_t = \tilde{X}^x \times_{\tilde{S}^s} t$. Then we say that $\tilde{X}^x_t$ is a variety of evanescent cycles of f at the point x.

We say that f is locally acyclic if, the reduced cohomology of every variety of evanescent cycles $\tilde{X}^x_t$ is trivial:

$$\tilde{H}^*(\tilde{X}^x_t,\mathbb{Z}/n) = 0, \tag{5.1.3.1}$$

i.e. $H^0(X^x_t,\mathbb{Z}/n) = \mathbb{Z}/n$ and $H^q(\tilde{X}^x_t,\mathbb{Z}/n) = 0$ for $q > 0$.

Lemma 5.1.4. *Let $f\colon X \longrightarrow S$ be a locally acyclic morphism and $g\colon S' \longrightarrow S$ a quasi-finite mophism (or the projective limit of quasi-finite morphisms). Then the morphism $f'\colon X' \longrightarrow S'$ derived from f by base-change is locally acyclic.*

We end up verifying that every variety of evanescent cycles of f' is a variety of evanescent cycles of f.

Lemma 5.1.5. *Let $f\colon X \longrightarrow S$ be a locally acyclic morphism. For every geometric point t of S, giving rise to a Cartesian diagram*

$$\begin{array}{ccc} X_t & \xrightarrow{\varepsilon'} & X \\ \downarrow & & \downarrow f \\ t & \xrightarrow{\varepsilon} & S \end{array}$$

We have $\varepsilon'_*\mathbb{Z}/n = f^*\varepsilon_*\mathbb{Z}/n$ and $R^q\varepsilon'_*\mathbb{Z}/n = 0$ for $q > 0$.

Let $\overline{S}$ be the closure of $\varepsilon(t)$, S' the normalization of $\tilde{S}$ in $k(t)$, and consider the Cartesian diagram:

$$\begin{array}{ccccc} X_t & \xrightarrow{i'} & X' & \xrightarrow{\alpha'} & X \\ \downarrow & & \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle f} \\ t & \xrightarrow{i} & S' & \xrightarrow{\alpha} & S \end{array}$$

The local rings of S' are normal to separably closed fraction fields. They are thus strictly Henselian, and the local acyclicity of f' (Lemma 5.1.4) provides $f'_*\mathbb{Z}/n = \mathbb{Z}/n$, $R^qf'_*\mathbb{Z}/n = 0$ for $q > 0$. Since α is an integer, we then have $R^q\varepsilon'_*\mathbb{Z}/n = \alpha'_*R^qi'_*\mathbb{Z}/n = \alpha'_*f'^*R^qi_*\mathbb{Z}/n = f^*\alpha_*R^qi_*\mathbb{Z}/n = f^*R^q\varepsilon_*\mathbb{Z}/n$; and hence the lemma.

5.1.6. Given a locally acyclic morphism $f\colon X \longrightarrow S$ and a specialization arrow $t \longrightarrow \tilde{S}^s$, we will define canonical homomorphisms, called cospecialization arrows

$$\mathrm{cosp}^*\colon H^*(X_t, \mathbb{Z}/n) \longrightarrow H^*(X_s, \mathbb{Z}/n),$$

connecting the cohomology of the general fiber $X_t = X \times_S t$ to that of the special fiber $X_s = X \times_S s$.

Consider the Cartesan diagram

$$\begin{array}{ccccc} X_t & \xrightarrow{\varepsilon'} & \tilde{X} & \longleftarrow & X_s \\ \downarrow & & \downarrow{\scriptstyle f'} & & \downarrow \\ t & \xrightarrow{\varepsilon} & \tilde{S}^s & \longleftarrow & s \end{array}$$

derived from f from base-change. By Lemma 5.1.4, f' is again locally acyclic. From the definition of local acyclicity, we immediately get that the restriction to X_s of the sheaves $R^q\varepsilon'_*\mathbb{Z}/n$ is $\mathbb{Z}/n$ for $q = 0$, and 0 for $q > 0$. By Lemma 5.1.5, we even know that $R^q\varepsilon'_*\mathbb{Z}/n = 0$ for $q > 0$. We define cosp^* as the composite arrow

$$H^*(X_t, \mathbb{Z}/n) \cong H^*(X, \varepsilon'_*\mathbb{Z}/n) \longrightarrow H^*(X_s, \mathbb{Z}/n) \tag{5.1.6.1}$$

Variant: Let $\overline{S}$ be the closure of $\varepsilon(t)$ in $\overline{S}^s$, S' the normalization of $\overline{S}$ in $k(t)$ and X'/S' derived from X/S by base-change. The diagram 5.1.6. can again be written as:

$$H^*(X_t, \mathbb{Z}/n) \cong H^*(X', \mathbb{Z}/n) \longrightarrow H^*(X_s, \mathbb{Z}/n).$$

Theorem 5.1.7. *Let S be a locally Noetherian scheme, s a geometric point of S and $f\colon X \longrightarrow S$ a morphism. Suppose*

(*a*) *the morphism f is locally acyclic,*

(*b*) *for every specialization morphism $t \longrightarrow \tilde{S}^s$ and for every $q \geqslant 0$, the cospecialization arrows $H^q(X_t, \mathbb{Z}/n) \longrightarrow H^q(X_s, \mathbb{Z}/n)$ are bijections.*

Then the canonical homomorphism $(R^qf_*\mathbb{Z}/n)_s \longrightarrow H^q(X_s, \mathbb{Z}/n)$ is bijective for every $q \geqslant 0$.

To prove the theorem, it is clear that we can suppose $S = \tilde{S}^s$. We will in fact show that, for every sheaf of $\mathbb{Z}/n\mathbb{Z}$-modules F on S, the canonical homomorphism $\varphi^q(F)\colon (R^qf_*f^*F)_s \longrightarrow H^q(X_s, f^*F)$ is bijective.

Every sheaf of $\mathbb{Z}/n\mathbb{Z}$-modules is the fitered inductive limit of constructible sheaves of $\mathbb{Z}/n\mathbb{Z}$-modules (Lemma 4.3.3). Moreover every constructible sheaf of $\mathbb{Z}/n\mathbb{Z}$-modules injects into a sheaf of the form $\prod i_{\lambda *}C_\lambda$, where $i_\lambda\colon t_\lambda \longrightarrow S$ is a finite family of generalizations of s and C_λ a free $\mathbb{Z}/n\mathbb{Z}$-module of finite rank over t_λ. By the definition of cospecialization arrows, condition (b) means that the homomorphisms $\varphi^q(F)$ are bijections if F is of this form.

We conclude using a variant of Lemma 4.3.6:

Lemma 5.1.6. *Let $\mathcal{C}$ be an abelian category in which filtered inductive limits exist. Let $\varphi^\bullet\colon T^\bullet \longrightarrow T'^\bullet$ be a morphism of cohomological functors commuting with filtered inductive limits, defined over $\mathcal{C}$ and with values in the category of abelian groups. Suppose that there exist two subsets $\mathcal{D}$ and $\mathcal{E}$ of objects of $\mathcal{C}$ such that:*

(*a*) *every object of $\mathcal{C}$ is a filtered inductive limit of objects in $\mathcal{D}$,*

(*b*) *every object of $\mathcal{D}$ is contained in an object of $\mathcal{E}$.*

Then the following conditions are equivalent.

(*i*) *$\varphi^q(A)$ is bijective for every $q \geqslant 0$ and every $A \in \mathrm{Ob}\,\mathcal{C}$.*

(*ii*) $\varphi^q(M)$ *is bijective for every* $q \geqslant 0$ *and every* $M \in \mathcal{E}$.

The proof of the lemma is accomplished by passing to the inductive limit, recurrence on q and repeated application of the five lemma of exact sequences in cohomology derived from an exact sequence $0 \longrightarrow A \longrightarrow M \longrightarrow A' \longrightarrow 0$, with $A \in \mathcal{D}$, $M \in \mathcal{E}$, $A' \in \mathrm{Ob}\,\mathcal{C}$.

Corollary 5.1.9. *Let S be the spectrum of a strictly henselian locally noetherian ring and* $f\colon X \longrightarrow S$ *a locally acyclic morphism. Suppose that, for every geometric point* t *of S we have* $H^0(X_t, \mathbb{Z}/n) = \mathbb{Z}/n$ *and* $H^q(X_t, \mathbb{Z}/n) = 0$ *for* $q > 0$ [*said otherwise the geometric fibers of f are acyclic*]. *Then we have* $f_*\mathbb{Z}/n = \mathbb{Z}/n$ *and* $R^q f_*\mathbb{Z}/n = 0$ *for* $q > 0$.

Corollary 5.1.10. *Let* $f\colon X \longrightarrow Y$ *and* $g\colon Y \longrightarrow Z$ *be morphisms of locally noetherian schemes. Then, if f and g are locally acyclic, then so is* $g \circ f$.

We can suppose that X, Y, and Z are strictly local and that f and g are local morphisms. It suffices then to show that, if z is an algebraic geometric point of Z, we have $H^0(X_z, \mathbb{Z}/n) = \mathbb{Z}/n$ and $H^q(X_z, \mathbb{Z}/n) = 0$ for $q > 0$.

Since g is locally acyclic, we have $H^0(Y_z, \mathbb{Z}/n) = \mathbb{Z}/n$, and $H^q(Y_z, \mathbb{Z}/n) = 0$ for $q > 0$. On the other hand, the morphism $f_z\colon X_z \longrightarrow Y_z$ is locally acyclic (Lemma 5.1.4) and its geometric fibers are acyclic, since these are the varieties of evanescent cycles of f. By Corollary 5.1.9, we then have $R^q f_{z*}\mathbb{Z}/n = 0$ for $q > 0$. Moreover $f_{z^*}\mathbb{Z}/n$ is constant on the fiber $\mathbb{Z}/n$ over Y_z. We are done with the help of the Leray spectral sequence for f_z.

5.2 Local acyclicity of a smooth morphism

Theorem 5.2.1. *A smooth morphism is locally acyclic.*

Let $f\colon X \longrightarrow S$ be a smooth morphism. The assertion is local for the étale topology on X and S, we can then suppose that X is an affine space of dimension d over S. Passing to the limit, we can suppose that S is Noetherian and the transitivity of local acyclicity (5.1.10) shows that it suffices to treat the case $d = 1$.

Let s be a geometric point of S and x a geometric point of X centered at a closed point of X_s. It suffices to show that the geometric fibers of the morphism $\tilde{X}^x \longrightarrow \tilde{S}^s$ are acyclic. We can write hereforth that $S = \tilde{S}^s = \mathrm{Spec}(A)$ and $X = \tilde{X}^x$. We have that $X \cong \mathrm{Spec}\{T\}$, from which we know that $A\{T\}$ is the Henselization of $A[T]$ at the point $T = 0$ over s.

If t is a geometric point of S, the fiber X_t is the projective limit of affine smooth curves over t. We then have $H^q(X_t, \mathbb{Z}/n) = 0$ for $q \geqslant 2$ and it suffices to show that $H^0(X_t, \mathbb{Z}/n) = \mathbb{Z}/n$ and $H^1(X_t, \mathbb{Z}/n) = 0$ for n coprime to the residual characteristic of S. This results in the following two propositions.

Proposition 5.2.2. *Let* A *be a strictly Henselian local ring,* $S = \mathrm{Spec}(A)$ *and* $X = \mathrm{Spec}\, A\{T\}$. *Then the geometric fibers of* $X \longrightarrow S$ *are connected.*

We can get here by passing to the limit to the case where A is a strict Henselization of a finite type $\mathbb{Z}$-algebra.

Let $\bar{t}$ be a geometric point of S, localized at t, and k' a finite separable extension of $k(t)$ in $k(\bar{t})$. We write $t' = \mathrm{Spec}(k')$ and $X_{t'} = X \times_S \mathrm{Spec}(k')$. It remains to verify that, for every $\bar{t}$ and t', $X_{t'}$ is connected (by which we mean connected and nonempty). Let A' be the normalization of A in k', i.e. the ring of elements of k' integers over the image of A in $k(t)$. We have $A\{T\} \times_A A' \xrightarrow{\sim} A'\{T\}$: the object on the left is in fact Henselian local (since A' is finite over A, and local) and the limit of étale local algebras over $A'[T] = A[T] \times_A A'$. The scheme X_t, is then again the fiber at t' of $X' = \mathrm{Spec}(A'\{T\})$ over $S' = \mathrm{Spec}(A')$. The local scheme X' is normal, thus integral; its localization X_t, is still integral, a fortiori connected.

Proposition 5.2.3. *Let A be a strict Henselian local ring, $S = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(A\{T\})$. Let $\bar{t}$ be a geometric point of S and $X_{\bar{t}}$ the corresponding geometric fiber. Then every étale galois cover of $X_{\bar{t}}$ of order coprime to the characteristic of the residual field of A is trivial.*

Lemma 5.2.3.1 (Zariski-Nagata theorem of purity in dimension 2)**.** Let C be a regular local ring of dimension 2 and C' an étale normal finite C-algebra over the open set complementary to the closed point of $\mathrm{Spec}(C)$. Then C' is étale over C.

In fact C' is normal of dimension 2, so $\mathrm{depth}(C') = 2$. Since $\mathrm{depth}(C') + \mathrm{pd}(C') = \dim(C) = 2$, we conclude that C' is free over C. So the set of points of C where C' is ramified is defined by an equation, the discriminant; since it doesn't contain a point of height 1, it is empty.

Lemma 5.2.3.2 (the particular case of Abhyankar's lemma)**.** Let $S = \mathrm{Spec}(V)$ be a line, π a uniformization, η the generic point of S, X smooth over S, irreducible, of relative dimension 1, $\tilde{X}_\eta$ a Galois étale covering of X_η, of degree n invertible over S, and $S_1 = \mathrm{Spec}(V[\pi^{1/n}])$. Note by an index $_1$ the base-change of S to S_1. Then, $\tilde{X}_{1\eta}$ extends to an étale covering of X_1.

Let $\tilde{X}_1$ be the normalization of X_1 in $\tilde{X}_{1\eta}$. Given the structure of the groups of moderate inertia of the localized valuation rings of X at the generic points of the special fiber X_s, $\tilde{X}_1$ is étale over X_1 over the general fiber, and at the generic points of the special fiber. By Lem. 5.2.3.1, it is étale everywhere.

5.2.3.3 We prove Prop. 5.2.3. Denote by t the point at which $\bar{t}$ is localized. It is acceptable to replace A by the normalization of A in a finite separable extension $k(t')$ of $k(t)$ in $k(\bar{t})$ (cf. Prop. 5.2.2). This, and passing to a preliminary limit, allows us to suppose that

(a) A is normal and Noetherian, and t is the generic point of S.

(b) The considered étale covering of $X_{\bar{t}}$ provides an étale covering of X_t.

(c) It provides an étale covering $\tilde{X}_U$ of the inverse image X_U of a nonempty open U of S (due to b): t is the limit of U).

(d) The complement of U is of codimension $\geqslant 2$ (this is at the cost of extending $k(t)$, by an application of 5.2.3.2 to the discrete valuation rings of S localized at the points of $S - U$ with codimension 1 in S; there are finitely many such points).

(e) The covering $\tilde{X}_U$ is trivial over the subscheme $T = 0$. (even if it means further extending $k(t)$).

When these conditions hold, we will see that the covering $\tilde{X}_U$ is trivial.

Lemma 5.2.3.4. Let A be a normal and Noetherian strictly henselian local ring, U an open in $\mathrm{Spec}(A)$ whose complement is of codimension $\geqslant 2$, V its inverse image in $X = \mathrm{Spec}(A\{T\})$ and V' an étale covering of V. If V' is trivial over $T = 0$, then V' is trivial.

Let $B = \Gamma(V', \mathcal{O})$. Since V' is the inverse image of V in $\mathrm{Spec}(B)$, it suffices to see that B is finite étale over $A\{T\}$ (so completely split, since $A\{T\}$ is strictly henselian). Let $\hat{X} = A[[T]]$, and let us observe by base changing from X to $\hat{X}$. The scheme $\hat{X}$ is faithfully flat over X. We thus have $\Gamma(\hat{V}', \mathcal{O}) = B \otimes_{A\{T\}} A[[T]]$, and it suffices to see that this ring $\hat{B}$ is finite étale over $A[[T]]$.

Let V_m (resp. V'_m) be the subscheme of $\hat{V}$ (respectively $\hat{V}'$) with the equation $T^{m+1} = 0$. By hypothesis, V'_0 is a trivial covering of V_0: a sum of n copies of V_0. Similarly for V'_m/V_m, since étale coverings are not sensitive to nilpotents. From this, we get

$$\varphi\colon \Gamma(\hat{V}', \mathcal{O}) \longrightarrow \varprojlim_m \Gamma(V'_m, \mathcal{O}) = (\varprojlim_m \Gamma(V_m, \mathcal{O}))^n$$

By hypothesis, the complement of U is of depth $\geqslant 2$: we have $\Gamma(V_m, \mathcal{O}) = A[T]/(T^{m+1})$, and φ is a homomorphism from $\hat{B}$ into $A[[T]]^n$. Over U, it furnishes n distinct sections of $\hat{V}'/\hat{V}$: $\hat{V}'$ is thus trivial, sum of n copies of $\hat{V}$. The complement of $\hat{V}$ in $\hat{X}$ is again of codimension $\geqslant 2$ (so of depth $\geqslant 2$), from which we deduce that $\hat{B} = A[[T]]^n$, and hence the lemma.

5.3 Applications

Theorem 5.3.1 (specialization of cohomology groups)**.** *Let $f\colon X \longrightarrow S$ be a proper and locally acyclic morphism, for example a smooth and proper morphism. Then the sheaves $R^q f_* \mathbb{Z}/n$ are locally constant constructible and for every specialization arrow $t \longrightarrow \tilde{S}^s$, the cospecialization arrows $H^q(X_t, \mathbb{Z}/n) \longrightarrow H^q(X_s, \mathbb{Z}/n)$ are bijective.*

This results immediately from the definition of cospecialization arrows and theorems of finiteness and basechange for proper morphisms.

Theorem 5.3.2 (basechange by a smooth morphism)**.** *Consider a cartesian diagram*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S \end{array}$$

with g smooth. For every sheaf F on X, of torsion coprime to the residual characteristic of S, we have

$$g^* R^q f_* F \xrightarrow{\sim} R^q f'_* (g'^* F).$$

Taking an open covering of X, we reduce to the case where X is affine, then by passing to the limit to the case where X is of finite type over S. Then f factorizes into an open immersion $j\colon X \longrightarrow \tilde{X}$ and a proper morphism $\tilde{f}\colon \tilde{X} \longrightarrow S$. From the Leray spectral sequence for $\tilde{f} = j$ and from the basechange theorem for proper morphisms, we deduce that it suffices to prove the theorem in the case where $X \longrightarrow S$ is an open immersion.

In this case, if F is of the form $\epsilon_* C$, where $\epsilon\colon t \longrightarrow X$ is a geometric point of X, the theorem is a corollary of 5.1.5. The general case follows by Lem. 5.1.6.

Corollary 5.3.3. *Let K/k be an extention of separably closed fields, X a k-scheme and n an integer coprime to the characteristic of k. Then the canonical map $H^q(X, \mathbb{Z}/n) \longrightarrow H^q(X_K, \mathbb{Z}/n)$ is bijectve for every $q \geqslant 0$.*

It suffices to remark that $\overline{K}$ is the inductive limit of smooth $\overline{k}$-algebras.

Theorem 5.3.4 (relative purity)**.** *Consider a commutative diagram:*

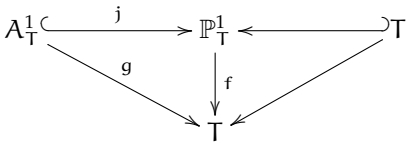
$$\begin{array}{ccccc} U & \xrightarrow{j} & X & \xleftarrow{i} & Y \\ & \searrow & \downarrow{\scriptstyle f} & \swarrow{\scriptstyle h} & \\ & & S & & \end{array} \qquad (5.3.4.1)$$

with f smooth of pure relative dimension N, h smooth of pure relative dimension $N-1$, i a closed submersion and $U = X - Y$. For n coprime to the residual characteristics of S, we have

$$\begin{cases} j_* \mathbb{Z}/n = \mathbb{Z}/n \\ R^1 j_* \mathbb{Z}/n = \mathbb{Z}/n(-1)_Y \\ R^q j_* \mathbb{Z}/n = 0 \qquad\qquad \textit{for } q \geqslant 2 \end{cases}$$

In these formulas, $\mathbb{Z}/n(-1)$ denotes the $\mathbb{Z}/n$-dual of μ_n. If t is a local equation for Y, the isomorphism $R^1 j_* \mathbb{Z}/n \simeq \mathbb{Z}/n(-1)_Y$ is defined by the map $a\colon \mathbb{Z}/n \longrightarrow R^1 j_* \mu_n$ which sends 1 onto the class of the μ_n-torsor of nth roots of t.

The question is of local nature. This allows us to replace (X, Y) by a locally isomorphic couple, for example



with $T = A^{n-1}_S$ and $i =$ the section at infinity. Cor. 5.1.9 applies to g, and yields $R^q g_* \mathbb{Z}/n = \mathbb{Z}/n$ for $q = 0$, 0 for $q > 0$. For f, we have on the other hand (Sec. 4 Eq. 6.2.1)

$$R^q f_* \mathbb{Z}/n = \mathbb{Z}/n,\ 0,\ \mathbb{Z}/n(-1),\ 0 \ \text{ for } \ q > 2.$$

We verify easily that $j_* \mathbb{Z}/n = \mathbb{Z}/n$, and that $R^q j_* \mathbb{Z}/n$ is concentrated on $i(T)$ for $q > 0$. The Leray spectral sequence $E_2^{p,q} = R^p f_* R^q j_* \mathbb{Z}/n \implies R^{p+q} g_* \mathbb{Z}/n$ therefore reduces to

$$\begin{array}{lccc} i^* R^q j_* \mathbb{Z}/n & 0 & \cdots & \\ i^* R^1 j_* \mathbb{Z}/n & 0 & \cdots & \\ \qquad\qquad \searrow^{d_2} & & & \\ \mathbb{Z}/n & 0 & \mathbb{Z}/n(-1) & 0 \quad \cdots \end{array}$$

$R^q j_* \mathbb{Z}/n = 0$ for $q \geqslant 2$, and $R^1 j_* \mathbb{Z}/n$ is the extension by zero of a locally free sheaf of rank one over T (isomorphic, via d_2 to $\mathbb{Z}/n(-1)$). Since the map a as defined above is injective (as we can check fiber by fiber), it is an isomorphism, and this proves Theorem 5.3.4.

5.3.5. We refer to [SGA4V3, XVI §4 and §5] for the proofs of the following applications of the acyclicity theorem (Thm. 5.2.1).

5.3.5.1. *Let $f: X \longrightarrow S$ be a morphism of finite type schemes over $\mathbb{C}$ and F a constructible sheaf on X. Then*

$$(R^q f_* F)^{an} \simeq R^q f^{an}_* (F^{an})$$

(cf. Thm. 4.6.3; in ordinary cohomology, it is necessary to suppose that F is constructible is not purely torsion).

5.3.5.2. *Let $f: X \longrightarrow S$ be a morphism of finite type schemes over a field of characteristic 0 and F a constructible sheaf on X. Then, $R^q f_* F$ is also constructible.*

The proof uses resolution of singularities and Theroem 5.3.4. It is generalized to the case of a finite type morphism of excellent schemes of characteristic 0 in [SGA4V3, XIX §5]. Another

proof, independent of resolution, is given in this volume (Sec. 1), it applies to a morphism of finite type schemes over a field or over a Dedekind ring.

6 Poincaré Duality

6.1 Introduction

Let X be an oriented topological manifold, of pure dimension N, and suppose that X admits a finite open cover $\mathcal{U} = (U_i)_{1 \leqslant i \leqslant K}$, such that the nonempty intersections of open subsets U_i are homeomorphic to balls. For such a variety, the theorem of Poincaré duality can be presented as follows.

A. The cohomology of X is the Ċech cohomology corresponding to the cover $\mathcal{U}$. That is the cohomology of the complex

$$0 \longrightarrow \mathbb{Z}^{A_0} \longrightarrow \mathbb{Z}^{A_1} \longrightarrow \cdots \tag{1}$$

where $A_k = \{(i_0, \ldots, i_k) | i_0 < \cdots < i_k \text{ and } U_i \cap \cdots \cap U_{i_k} \neq \emptyset\}$.

B. For $a \in A_k$, $a = (i_0, \ldots, i_k)$, let $U_a = U_{i_0} \cap \ldots U_{i_k}$ and let j_a be the inclusion of U_a into X. The constant sheaf $\mathbb{Z}$ on X admits a resolution (to the left)

$$\cdots \longrightarrow \bigoplus_{a \in A_1} j_{a!}\mathbb{Z} \longrightarrow \bigoplus_{a \in A_0} j_{a!}\mathbb{Z} \longrightarrow 0 \tag{2}$$
$$\downarrow$$
$$\mathbb{Z}$$

Cohomology with compact support $H^*_c(X, j_{a!}\mathbb{Z})$ is nothing but the cohomology with proper support of the (oriented) ball U_a:

$$H^i_c(X, j_{a!}\mathbb{Z}) = \begin{cases} 0 & \text{if } i \neq N \\ \mathbb{Z} & \text{if } i = N \end{cases}$$

The hypercohomology spectral sequence, for the complex 2 and the cohomology with proper support, confirms thus that $H^i_c(X)$ is the $(i-N)$th cohomology group of a complex

$$\cdots \longrightarrow \mathbb{Z}^{A_1} \longrightarrow \mathbb{Z}^{A_0} \longrightarrow 0. \tag{3}$$

This complex is the dual of the complex 1, whence Poincaré duality.

The essential points of this construction are a) the existence of a theory of cohomology with proper support: b) the fact that every point x of a manifold

X of pure dimension N has a fundamental system of open neighborhoods U for which

$$H_c^i(U) = \begin{cases} 0 & \text{for } i \neq N, \\ \mathbb{Z} & \text{for } i = N \end{cases} \tag{4}$$

Poincaré duality in étale cohomology can be constructed on this model. For X smooth of pure dimension N over k algebraically closed, n invertible over X and x a closed point of X, the key point is to calculate the projective limit, taken over the étale neighborhoods U of x.

$$\varprojlim H_c^i(U, \mathbb{Z}/n) = \begin{cases} 0 & \text{if } i \neq 2N \\ \mathbb{Z}/n & \text{if } i = 2N. \end{cases} \tag{5}$$

Similarly for topological manifolds one must first directly treat the case of an open ball (or even simply that of the interval $]0,1[$), here one must first directly treat the case of curves (§6.2). The local acyclicity theorem for smooth morphisms allows then to bring 5 to this particular case (§6.3).

The isomorphisms 4 and 5 are not canonical: they depend on the choice of an *orientation* of X. For n invertible on a scheme X, μ_n is a sheaf of $\mathbb{Z}/n$-modules of rank one. We denote by $\mathbb{Z}/n(N)$ its Nth tensor power ($N \in \mathbb{Z}$). The intrinsic form of the second line of 5 is

$$\varprojlim H_c^{2N}(U, \mathbb{Z}/n(N)) = \mathbb{Z}/n, \tag{5'}$$

and $\mathbb{Z}/n(N)$ is called the *orientation sheaf* of X; the sheaf $\mathbb{Z}/n(N)$ being constant, isomorphic to $\mathbb{Z}/n$, we can bring it out of the symbol H and rather write

$$\varprojlim H_c^{2N}(U, \mathbb{Z}/n) = \mathbb{Z}/n(-N), \tag{5''}$$

and Poincaré duality takes the form of a perfect duality, with values in $\mathbb{Z}/n(-N)$, between $H^i(X, \mathbb{Z}/n)$ and $H_c^{2N-i}(X, \mathbb{Z}/n)$.

6.2 The case of curves

6.2.1. Let $\overline{X}$ be a smooth projective curve over an algebraically closed k, and n invertible on $\overline{X}$. The proof of 3.3.5 yields, for $\overline{X}$ connected, a canonical isomorphism

$$H^2(\overline{X}, \mu_n) = \mathrm{Pic}(\overline{X})/n\,\mathrm{Pic}(\overline{X}) \xrightarrow[\sim]{\deg} \mathbb{Z}/n$$

Let D be a reduced divisor of $\overline{X}$ and $X = \overline{X} - D$.

$$X \xhookrightarrow[j]{} \overline{X} \xhookleftarrow[i]{} D$$

The exact sequence $0 \longrightarrow j_!\mu_n \longrightarrow \mu_n \longrightarrow i_*\mu_n \longrightarrow 0$ and the fact that $H^i(\overline{X}, i_*\mu_n) = H^i(D, \mu_n) = 0$ for $i > 0$ yield an isomorphism

$$H^2_c(X, \mu_n) = H^2(\overline{X}, j_!\mu_n) \xrightarrow{\sim} H^2(\overline{X}, \mu_n) \xrightarrow{\sim} \mathbb{Z}/n.$$

For $\overline{X}$ disconnected, we also have

$$H^2_c(X, \mu_n) \simeq (\mathbb{Z}/n)^{\pi_0(X)}$$

and we define the *trace morphism* as the sum

$$\mathrm{Tr}\colon H^2_c(X, \mu_n) \simeq (\mathbb{Z}/n)^{\pi_0(X)} \xrightarrow{\Sigma} \mathbb{Z}/n$$

Theorem 6.2.2. *The form* $\mathrm{Tr}(a \cup b)$ *identifies each of the groups* $H^i(X, \mathbb{Z}/n)$ *and* $H^{2-i}_c(X, \mu_n)$ *with the dual (with respect to* $\mathbb{Z}/n$*) of the other.*

(*Transcendental proof*) If $\overline{X}$ is a smooth projective curve on a trait S and that $j\colon X \hookrightarrow \overline{X}$ is the inclusion of the complement of an étale divisor D on S, the cohomologies (resp. the cohomologies with proper support) of the generic and special geometric fibers of X/S are "the same," i.e. the fibers of locally constant sheaves on S. This can be deduces from the analogous facts for $\overline{X}$ and D, via the exact sequence $0 \longrightarrow j_!\mathbb{Z}/n \longrightarrow \mathbb{Z}/n \longrightarrow \mathbb{Z}/n_D \longrightarrow 0$ (for cohomology with proper support) and the formulas $j_*\mathbb{Z}/n = \mathbb{Z}/n$, $R^1j_*\mathbb{Z}/n \simeq \mathbb{Z}/n_D(-1)$, $R^ij_*\mathbb{Z}/n = 0\,(i \geqslant 2)$ (for ordinary cohomology) (Thm. 5.3.4).

This principle of specialization reduces Thm. 6.2.2 to the case where k is of characteristic 0. By Cor. 5.3.3, this case reduces to the one with $k = \mathbb{C}$. Finally, for $k = \mathbb{C}$, the groups $H^*(X, \mathbb{Z}/n)$ and $H^*_c(X, \mu_n)$ coincide with their namesake groups, calculated for the classical topological space X_{cl} and, via the isomorphism $\mathbb{Z}/n \longrightarrow \mu_n\colon X \mapsto \exp\left(\frac{2\pi i x}{n}\right)$, the trace morphism is identified with "integration on the fundamental class," so that 6.2.2 results from Poincaré duality for X_{cl}.

6.2.3. (*Algebraic proof*) For a very economical proof, see §2. Here is another one, tied to the autoduality of the Jacobian.
Let's continue with the notation from 6.2.1. We can suppose - and we will suppose - that X is connected. The case $i = 0$ or 2 being trivial, we also suppose that $i = 1$. Let us define ${}_D\mathbb{G}_m$ by the exact sequence $0 \longrightarrow_D \mathbb{G}_m \longrightarrow \mathbb{G}_m \longrightarrow i_*\mathbb{G}_m \longrightarrow 0$ (sections of $\mathbb{G}_m$ congruent to 1 mod D). The group $H^1(\overline{X}, {}_D\mathbb{G}_m)$ classifies invertible sheaves on $\overline{X}$ trivialized over D. This is the group of points of $\mathrm{Pic}_D(\overline{X})$, an extension of $\mathbb{Z}$ (the degree) by the group of points of a generalized Jacobian of Rosenlicht (corresponding to the conductor 1 in each point of D) $\mathrm{Pic}^0_D(\overline{X})$, itself an extension of the abelian variety $\mathrm{Pic}^0(\overline{X})$ by the torus $\mathbb{G}_m^D/(\mathbb{G}_m$ diagonal$)$.

a) The exact sequence $0 \longrightarrow j_!\mu_n \longrightarrow {}_D\mathbb{G}_m \xrightarrow{x\mapsto x^n} {}_D\mathbb{G}_m \longrightarrow 0$ furnishes an isomorphism

$$H^1_c(X, \mu_n) = \mathrm{Pic}^0_D(\overline{X})_n. \tag{6.2.3.1}$$

b) The map which sends $x \in X(k)$ to the class of the invertible sheaf $\mathcal{O}(x)$ over $\overline{X}$, trivialized by 1 over D, provides a morphism

$$f\colon X \longrightarrow \mathrm{Pic}_D(\overline{X}).$$

For the following, we fix a basepoint 0 and write $f_0(x) = f(x) - f(0)$. for every homomorphism $\nu\colon \mathrm{Pic}^0_D(\overline{X})_n \longrightarrow \mathbb{Z}/n$, let $\overline{v} \in H^1(\mathrm{Pic}^0_D(\overline{X}), \mathbb{Z}/n)$ be the image under ν of the class in $H^1(\mathrm{Pic}^0_D(\overline{X}), \mathrm{Pic}^0_D(\overline{X})_n)$ of the torsor defined by the extension

$$0 \longrightarrow \mathrm{Pic}^0_D(\overline{X})_n \longrightarrow \mathrm{Pic}^0_D(\overline{X}) \xrightarrow{n} \mathrm{Pic}^0_D(\overline{X}) \longrightarrow 0$$

Geometric class field theory (as presented in [Ser59]) shows that the map $\nu \mapsto f_0^*(\overline{v})$:

$$\mathrm{Hom}(\mathrm{Pic}^0_D(\overline{X})_n, \mathbb{Z}/n) \longrightarrow H^1(X, \mathbb{Z}/n) \tag{6.2.3.2}$$

is an isomorphism. To deduce Thm. 6.2.2 from 6.2.3.1 and 6.2.3.2, it remains to know that

$$\mathrm{Tr}(u \cup f_0^*(\overline{v})) = -\nu(u). \tag{6.2.3.3}$$

This compatibility is proven in 3.

6.3 The general case

Let X be a smooth algebraic variety of pure dimension N, over k algebraically closed. To state the Poincaré duality theorem, one must first define the trace morphism

$$\mathrm{Tr}\colon H^{2N}_c(X, \mathbb{Z}/n(N)) \longrightarrow \mathbb{Z}/n.$$

The definition is a hard dévissage beginning with the case of curves ([SGA4V3, XVIII §2]). We thus have

Theorem 6.3.1. *The form* $\mathrm{Tr}(a \cup b)$ *identifies each of the groups* $H^i_c(X, \mathbb{Z}/n(N))$ *and* $H^{2N-i}(X, \mathbb{Z}/n)$ *with the dual of the other.*

Let $x \in X$ be a closed point and X_x the strict localization of X at x. We set, for U ranging over the étale neighborhoods of x

$$H^*_c(X_x, \mathbb{Z}/n) = \varprojlim H^*_c(U, \mathbb{Z}/n). \tag{1}$$

It will be preferable to rather consider the pro-object "$\varprojlim$" $H_c^*(U, \mathbb{Z}/n)$ but, with the groups involved being finite, the distinction does not matter. As we have tried to explain in the introduction, Thm. 6.3.1 results from the fact that

$$\begin{aligned} &H_c^i(X_x, \mathbb{Z}/n) = 0 \text{ for } i \neq 2N \text{ and that} \\ &\mathrm{Tr}\colon H_c^{2N}(X_x, \mathbb{Z}/n(N)) \xrightarrow{\sim} \mathbb{Z}/n \text{ is an isomorphism.} \end{aligned} \tag{2}$$

The case where $N = 0$ is trivial. If $N > 0$, let Y_y be the strict localization at a closed point of a smooth scheme Y of pure dimension $N - 1$, and let $f\colon X_x \longrightarrow Y_y$ be an essentially smooth morphism (of relative dimension one). The proof uses the Leray spectral sequence in cohomology with proper support for f, to reduce to the case of curves. The "cohomologies with proper support" considered being defined as limits 1, the existence of such a spectral sequence poses various problems of passing to the limit, treated with too much detail in [SGA4V3, p. XVIII]. Here, we will content ourselves by calculating.

For every geometric point z of Y_y, we have

$$(R^i f_! \mathbb{Z}/n)_z = H_c^i(f^{-1}(z), \mathbb{Z}/n).$$

The geometric fiber $f^{-1}(z)$ is a projective limit of smooth curves on an algebraically closed fieled. It satisfies Poincaré duality. Its ordinary cohomology is given by the local acyclicity theorem for smooth morphisms:

$$H^i(f^{-1}(z), \mathbb{Z}/n) = \begin{cases} \mathbb{Z}/n & \text{for } i = 0 \\ 0 & \text{for } i = 0. \end{cases}$$

By duality, we have

$$H_c^i(f^{-1}(z), \mathbb{Z}/n) = \begin{cases} \mathbb{Z}/n(-1) & \text{for } i = 2 \\ 0 & \text{for } i \neq 2 \end{cases}$$

and the Leray spectral sequence can be written as

$$H_c^i(X_x, \mathbb{Z}/n(N)) = H_c^{i-2}(Y_y, \mathbb{Z}/n(N-1)).$$

We conclude by recurrence on N.

6.4 Variants and Applications

We can construct, in étale cohomology, a "duality formalism" (= of the functors Rf_*, $Rf_!$, f^*, $Rf^!$ satisfying various compatibilities and adjunction formulas) parallel to that which exists in coherent cohomology. In this language, the results of the preceding paragraphs can be transcribed as follows: if $f\colon X \longrightarrow S$ is smooth of pure relative dimension N, and $S = \mathrm{Spec}(k)$ with k algebraically closed, then

$$Rf^! \mathbb{Z}/n = \mathbb{Z}/n[2N](N).$$

This result holds with no hypotheses on S. It follows that

Corollary 6.4.1. *If f is smooth and of pure relative dimension N, and the sheaves $R^i f_! \mathbb{Z}/n$ are locally constant, then the sheaves $R^i f_* \mathbb{Z}/n$ are also locally constant, and*

$$R^i f_* \mathbb{Z}/n = \underline{\mathrm{Hom}}(R^{2N-i} f_! \mathbb{Z}/n(N), \mathbb{Z}/n).$$

In particular, under the hypotheses of the corollary, the sheaves $R^i f_* \mathbb{Z}/n$ are constructible. Starting there, we can show that, if S is of finite type over the spectrum of a field or of a Dedekind ring, then, for every finite type morphism $f\colon X \longrightarrow S$ and every constructible sheaf $\mathcal{F}$ on X, the sheaves $R^i f_* \mathcal{F}$ on S are constructible (Th. finitude, 1.1).

References

[Art62] M. Artin. *Grothendieck Topologies: Notes on a Seminar*. Harvard University, Department of Mathematics, 1962. URL: https://books.google.co.in/books?id=wERAAQAAIAAJ.

[Art69] M. Artin. "Algebraic approximation of structures over complete local rings". In: *Inst. Hautes Études Sci. Publ. Math.* 36 (1969), pp. 23–58. ISSN: 0073-8301,1618-1913. URL: http://www.numdam.org/item?id=PMIHES_1969__36__23_0.

[Art73] Michael Artin. *Théorèmes de représentabilité pour les espaces algébriques*. Séminaire de Mathématiques Supérieures 44. In collaboration with Alexandru Lascu and Jean-François Boutot. Montréal, Qué.: Les Presses de l'Université de Montréal, 1973. ISBN: 0840502192.

[EGA3] A. Grothendieck. "Éléments de géométrie algébrique. III. Étude cohomologique des faisceaux cohérents. I." In: *Inst. Hautes Études Sci. Publ. Math.* 11 (1961), p. 167. ISSN: 0073-8301,1618-1913. URL: http://www.numdam.org/item?id=PMIHES_1961__11__167_0.

[Gir64] Jean Giraud. "Analysis situs". In: *Séminaire Bourbaki, 1962/63. Fasc. 3*. Secrétariat mathématique, Paris, 1964, No. 256, 11.

[Nag62] Masayoshi Nagata. "Imbedding of an abstract variety in a complete variety". In: *Journal of Mathematics of Kyoto University* 2.1 (1962), pp. 1–10. DOI: 10.1215/kjm/1250524969. URL: https://projecteuclid.org/journals/kyoto-journal-of-mathematics/volume-2/issue-1/Imbedding-of-an-abstract-variety-in-a-complete-variety/10.1215/kjm/1250524969.full.

[Nag63] Masayoshi Nagata. "A generalization of the imbedding problem of an abstract variety in a complete variety". In: *Journal of Mathematics of Kyoto University* 3.1 (1963), pp. 89–102. DOI: 10.1215/kjm/1250524859. URL: https://projecteuclid.org/journals/kyoto-journal-of-mathematics/volume-3/issue-1/A-generalization-of-the-imbedding-problem-of-an-abstract-variety/10.1215/kjm/1250524859.full.

[Ray70] Michel Raynaud. *Anneaux locaux henséliens*. Vol. Vol. 169. Lecture Notes in Mathematics. Springer-Verlag, Berlin-New York, 1970, pp. v+129.

[Ser56] Jean-Pierre Serre. "Géométrie algébrique et géométrie analytique". In: *Annales de l'Institut Fourier* 6 (1956), pp. 1–42. URL: https://www.numdam.org/item/AIF_1956__6__1_0/.

[Ser59] Jean-Pierre Serre. *Groupes algébriques et corps de classes*. Vol. 1264. Actualités scientifiques et industrielles. Cours au Collège de France. Paris: Hermann, 1959.

[Ser62] Jean-Pierre Serre. *Corps locaux*. Vol. VIII. Publications de l'Institut de Mathématique de l'Université de Nancago. Actualités Scientifiques et Industrielles, No. 1296. [Current Scientific and Industrial Topics]. Hermann, Paris, 1962, p. 243.

[Ser65] Jean-Pierre Serre. *Cohomologie galoisienne*. Vol. No. 5. Lecture Notes in Mathematics. With a contribution by Jean-Louis Verdier, Troisième édition, 1965. Springer-Verlag, Berlin-New York, 1965, v+212 pp. (not consecutively paged).

[SGA1] *Revêtements étales et groupe fondamental*. Vol. Vol. 224. Lecture Notes in Mathematics. Séminaire de Géométrie Algébrique du Bois Marie 1960–1961 (SGA 1), Dirigé par Alexandre Grothendieck. Augmenté de deux exposés de M. Raynaud. Springer-Verlag, Berlin-New York, 1971, pp. xxii+447.

[SGA4V3] *Théorie des topos et cohomologie étale des schémas. Tome 3*. Vol. Vol. 305. Lecture Notes in Mathematics. Séminaire de Géométrie Algébrique du Bois-Marie 1963–1964 (SGA 4), Dirigé par M. Artin, A. Grothendieck et J. L. Verdier. Avec la collaboration de P. Deligne et B. Saint-Donat. Springer-Verlag, Berlin-New York, 1973, pp. vi+640.

[Tse33] Ch. C. Tsen. "Divisionsalgebren über Funktionenkörpern". In: *Nachrichten von der Gesellschaft der Wissenschaften zu Göttingen, Mathematisch-Physikalische Klasse* 1933 (1933), pp. 335–339. URL: http://eudml.org/doc/59436.

Chapter 3

Report on the trace formula

In this text, I've tried to present Grothendieck's cohomological theory of L-functions as directly as possible. I am following very closely, some lectures given by Grothendieck at the IHES in the spring of 1966. In the spirit of this volume, no appeal will be made to SGA 5 - except two references to passages from Exposé XV, which is independent from the rest of this seminar.

In §10 (p.81-90) of [Del73], the reader will find a summary of the theory, in the crucial case of curves.

Contents

1 Some reminders 63
2 $\mathbb{Q}_\ell$-sheaves 64
3 Trace formula and L-functions 64
4 Reductions to theorems in $\mathbb{Z}/\ell^n$ cohomology 64
5 The Nielsen-Wicken method 64
6 End of the proof of Thm. 3.4.10 64

---

1 Some reminders

1.1 Notations. p, q, $\mathbb{F}$, $\mathbb{F}_q$: p is a prime number, $q = p^f$ a power of p and $\mathbb{F}$ an algebraic closure of the prime field $\mathbb{F}_p$. For every power p^n of p, we denote by $\mathbb{F}_{p^n}$ the subfield with p^n elements of $\mathbb{F}$.

X_0, X: We will often designate by a symbol with an index 0 an object over $\mathbb{F}_q$. The same symbol, without 0, will denote the object derived from it by extending scalars to $\mathbb{F}$. For example, if $\mathcal{F}_0$ is a sheaf on a scheme X_0 over $\mathbb{F}_q$, we denote by $\mathcal{F}$ its inverse image over $X = X_0 \otimes_{\mathbb{F}_q} \mathbb{F}$.

F: If X_0 is a scheme over $\mathbb{F}_q$, we will term *Frobenius morphism*, and denote by F, the endomorphism of X_0 which "sends the point with coordinates x to the point with coordinates x^q": it is the identity on underlying topological spaces and, for x a local section of the structure sheaf, $F^*x = x^q$.

We will denote again by F the endomorphism of X obtained by extension of scalars. On the set $X(\mathbb{F}) = X_0(\mathbb{F})$ of rational points of X, F acts as the Frobenius substitution $\varphi \in \mathrm{Gal}(\mathbb{F}/\mathbb{F}_q)$ defined by $\varphi(x) = x^q$. In particular $X^F = X_0(\mathbb{F}_q)$.

In cases with ambiguity, we will write $F_{(q)}$ rather than F.

2 $\mathbb{Q}_\ell$-sheaves

3 Trace formula and L-functions

4 Reductions to theorems in $\mathbb{Z}/\ell^n$ cohomology

5 The Nielsen-Wicken method

6 End of the proof of Thm. 3.4.10

Chapter 4

L-functions modulo ℓ^n and modulo p

Contents

1 Symmetric tensors 65
2 The theorem 65
3 Artin-Scheier Theory 65
4 Trace formula modulo p 65

---

1 Symmetric tensors

2 The theorem

3 Artin-Scheier Theory

4 Trace formula modulo p

Chapter 5

The cohomology class associated to a cycle

Contents

1 Cohomology with support and cup-products 68

1.1 H^1 and torsors 68

1.2 Cup-products 68

1.3 The Koszul rule 68

2 The cohomology class associated to a cycle 68

2.1 The class of a divisor 68

2.2 Cohomological method 68

2.3 Homological method 68

3 Application : the Lefschetz trace formula in the proper and smooth case 68

1 Cohomology with support and cup-products

1.1 H^1 **and torsors**

1.2 Cup-products

1.3 The Koszul rule

2 The cohomology class associated to a cycle

2.1 The class of a divisor

2.2 Cohomological method

2.3 Homological method

3 Application : the Lefschetz trace formula in the proper and smooth case

Chapter 6

Duality

Contents

1 Local biduality, in dimension 1 69
2 Poincaré duality for curves, after M. Artin 69
3 Poincaré duality and the Jacobian 69
4 Compatibility SGA 4 XVIII 3.1.10.3 69

---

1 Local biduality, in dimension 1

2 Poincaré duality for curves, after M. Artin

3 Poincaré duality and the Jacobian

4 Compatibility SGA 4 XVIII 3.1.10.3

Chapter 7

Applications of the trace formula to trigonometric sums

Contents

1 Principles 72

2 Dictionary 72

3 Sums in one variable 72

4 Gauss sums and Jacobi sums 72

5 Hecke characters 72

6 Hecke characters defined by Jacobi sums 72

7 Generalized Kloosterman sums 72

8 Other applications 72

1 Principles

2 Dictionary

3 Sums in one variable

4 Gauss sums and Jacobi sums

5 Hecke characters

6 Hecke characters defined by Jacobi sums

7 Generalized Kloosterman sums

8 Other applications

Chapter 8

Theorems of finiteness in ℓ-adic cohomology

Contents

1 Statement of the theorems 73
2 Generic theorems 73
3 Proof of ... and constructibility of sheaves of evanescent cycles 73
4 Local biduality 73
5 Appendix 73
5.1 Cohomological property 73
5.2 Specialization and cospecialization 73

---

1 Statement of the theorems

2 Generic theorems

3 Proof of ... and constructibility of sheaves of evanescent cycles

4 Local biduality

5 Appendix

5.1 Cohomological property

5.2 Specialization and cospecialization

Chapter 9

Derived categories, état 0

Contents

1 Triangulated categories 76

1.1 Definitions and Examples 76

1.2 Quotient categories 76

2 Applications to abelian categories 76

2.1 Derived categories 76

2.2 Derived functors 76

2.3 Examples 76

1 Triangulated categories

1.1 Definitions and Examples

1.1.1 Definition of triangulated categories. First properties

1.1.2 Examples of triangulated categories

1.1.3 Examples of exact functors and of cohomological functors

1.2 Quotient categories

1.2.1 Thick categories. Multiplicative systems

1.2.2 Universal equivalent problems

1.2.3 Calculus of fractions

1.2.4 Properties of quotient categories

1.2.5 Properties of the functor of passing to the quotient

1.2.6 Orthogonality

2 Applications to abelian categories

2.1 Derived categories

2.1.1 The derived categories of an abelian category

2.1.2 Study of Ext

2.2 Derived functors

2.2.1 Definition of derived functors

2.2.2 Existence of derived functors

2.2.3 Cup-product. Composition

2.3 Examples

2.3.1 The Hom functor

2.3.2 Tor of sheaves

2.3.3 The left derived functor of the inverse image

Chapter 10

Erratum for SGA 4